Data Science Project, Phase 1

Clean the Data, Explore It, and Set a Baseline

Introduction to Programming with Python

Group tasks built on Section 7.1 and Sections 11.1 to 11.4

27 tasks, one for each group of students

How This Phase Fits the Final Project

In the final project you will train machine learning models. A model learns only from the table it is given. If the table has copies, holes and typing errors, the model learns those too.

In this phase your group gets one dataset. You clean it, explore it, and draw it. Then you measure two simple predictions on it. The first prediction ignores every column. The second is a straight-line fit. The models of the final project must beat both numbers.

Each task names a **target**: the column your model will predict. In a **regression** task the target is a number, such as a rent. In a **classification** task the target is a label, such as Yes or No.

The data of every task are invented for practice. The patterns in them follow the situation the task describes.

Group Allocation

Each group does the task with its own number. Click a task to go to its page.

Group	Task	Target	Kind
1	Apartment Rents in Karachi	`monthly_rent`	regression
2	Used Car Prices in Lahore	`price`	regression
3	Final Results at a Multan College	`result`	classification
4	Wheat Yield on Punjab Farms	`yield_maunds_per_acre`	regression
5	Customers Leaving a Telecom Company in Islamabad	`churned`	classification
6	Small Loans in Faisalabad	`defaulted`	classification
7	Food Delivery Times in Peshawar	`delivery_minutes`	regression
8	Diabetes Screening Camps near Hyderabad	`diabetic`	classification
9	Export Grade of Kinnow from Sargodha	`export_grade`	classification
10	Unsafe Well Water in Quetta	`unsafe`	classification
11	Faulty Footballs in Sialkot	`defective`	classification
12	Returned Orders at an Online Clothing Store	`returned`	classification
13	Electricity Used by Steel Furnaces in Gujranwala	`energy_kwh`	regression
14	Cancelled Bookings at Northern Guesthouses	`cancelled`	classification
15	Daily Output of Solar Systems in Bahawalpur	`daily_kwh`	regression
16	Traffic at Motorway Toll Plazas	`vehicles_per_hour`	regression
17	Milk Yield of Buffaloes in Pakpattan	`milk_litres`	regression
18	Delayed Flights out of Karachi	`delayed`	classification

Group	Task	Target	Kind
19	Resale Prices of Used Smartphones	`resale_price`	regression
20	Staff Leaving a Lahore Software House	`left_company`	classification
21	Sugar Recovery at Mills in Sindh	`recovery_pct`	regression
22	Children Leaving School in Tharparkar	`dropped_out`	classification
23	Restaurant Ratings in Islamabad and Rawalpindi	`average_rating`	regression
24	Fraud on a Mobile Wallet in Sindh	`fraud`	classification
25	Groundwater Depth in Central Punjab	`water_depth_ft`	regression
26	Gym Membership Renewals in Rawalpindi	`renewed`	classification
27	Container Clearance Time at Karachi Port	`clearance_hours`	regression

Files and Software

Every task has two files in the folder `Project_Phase1_Data`:

- `taskNN_<name>.csv`, the main table, where `NN` is your group number;
- `taskNN_<name>_lookup.csv`, a small table with one row for each value of one column.

Never edit a data file by hand. Every fix must be made by your program, so that anyone can run it again and get the same table.

Install the libraries once from the command line:

```
pip install numpy pandas matplotlib seaborn
```

The check values in this sheet were produced with Python 3.11.9, NumPy 2.4.6, pandas 3.0.5, Matplotlib 3.11.1 and Seaborn 0.13.2. Use only the Python of the book: Section 7.1, Sections 11.1 to 11.4, and the chapters before them. One small step goes beyond Section 11.4. There, `.values` turned one column into a 1-D array. On several columns, `clean[["a", "b"]].values` gives a 2-D array with one row for each row of the table.

What Every Group Does

The parts below are the same for every group. Your task page gives the column names, the cleaning rules, the questions and the check values for your dataset. Put all your work in one program or notebook.

Part A: Load and Inspect (Section 11.2)

A1. Read both files with `pd.read_csv`. Print the shape and the first rows of each.

A2. Print the column types with `info()`, the summary with `describe()`, and the missing values of each column with `isna().sum()`.

A3. Count the rows that are exact copies with `duplicated().sum()`. Print `value_counts()` of the column that is the merge key.

A4. Write down every problem you can see before you clean anything: copies, missing values, impossible numbers, text where numbers should be, and names spelled in more than one way.

Part B: Clean (Sections 11.2 and 11.3)

Follow the cleaning rules on your task page in the order given. These tools do the work: `drop_duplicates`, `.str.strip` and `.str.title`, `.str.replace` with `pd.to_numeric`, `pd.to_datetime`, a mask with `~` and `isna`, `median` with `fillna`, and `merge` with `how="left"`. Print the number of rows after each step and compare it with the check values. A different number means a step went wrong, or the steps ran in another order.

Part C: Explore with Pandas (Section 11.3)

Answer the five questions on your task page. Use `groupby` with `agg`, `pivot_table`, the `.dt.month` of the date column, and masks. Give each answer as a number, rounded to two or three decimal places, and one sentence that says what it means.

Part D: Arrays and a Baseline (Sections 7.1 and 11.1)

D1. Build the feature array `X` and the target array `y` from the columns named on your task page. Print their shapes.

D2. Print the mean and the standard deviation of each column of `X` with `axis=0`. Turn `X` into z-scores by broadcasting. Confirm with `np.allclose` that every column of z-scores has mean 0.

D3. Find the correlation of each feature with `y` using `np.corrcoef`. Sort the features by the size of the correlation, ignoring its sign. Name the strongest three.

D4. Split the rows into a training part and a test part, train the two baseline predictions, and measure them on the test part. Use the code below with your own names and seed.

D5. Compare the two numbers. Say in one sentence what the straight-line fit gained over the baseline.

Part E: Charts (Section 7.1 and Section 11.4)

Give every chart a title and labelled axes with units. Your task page names the columns to use.

E1. The distribution of the target, coloured by the category column.

E2. The target across the groups of the merge key.

E3. The target against your strongest feature from step D3: a scatter plot coloured by the category column for regression, or a box plot of the feature for each label for classification.

E4. A heatmap of the correlations between the features and the target, with the values written in the cells.

E5. One figure-level chart (`sns.relplot`, `sns.catplot` or `sns.displot`) with one panel for each value of the category column.

E6. A dashboard: charts E1 to E4 on one Figure of 2 by 2 Axes, saved with `fig.savefig` as `groupNN_dashboard.png` at `dpi=150`.

E7. An outlier check: compute the quartiles with `quantile`, then the fences 1.5 times the IQR beyond them. Count the values outside the fences, and say whether you would keep them.

Part F: Save and Report

Save the cleaned and merged table with `to_csv` as `groupNN_clean.csv`, with `index=False`. The final project starts from this file. Then write a report of at most two pages. It must contain:

- the problems you found in Part A, and how many rows each cleaning step changed;
- a statement that your check values match the task page, or where and why they differ;
- three findings from Part C, each with its number;
- two of your charts, each with one sentence saying what it shows;
- the three strongest features from step D3, and the two baseline numbers from step D4;
- one paragraph on the final project: which columns a model should use, and which number it must beat.

The Code for Part D

Part D uses the same code in every task. Only the column names and the seed change. The code below is filled in for Task 1, a regression task. Its printout matches the check values on the Task 1 page.

```
import numpy as np

# clean is your cleaned and merged table from Part B.
features = [
    "size_sqft", "bedrooms", "floor", "building_age_years",
    "distance_to_saddar_km",
]
X = clean[features].values
y = clean["monthly_rent"].values

rng = np.random.default_rng(2001)
in_train = rng.random(len(clean)) < 0.8
X_train, X_test = X[in_train], X[~in_train]
y_train, y_test = y[in_train], y[~in_train]
print("Train rows:", len(y_train), " Test rows:", len(y_test))

# Baseline: predict the training mean for every test row.
baseline = np.full(len(y_test), y_train.mean())
baseline_rmse = np.sqrt(np.mean((y_test - baseline) ** 2))
print("Baseline RMSE:", round(baseline_rmse, 2))

# Standardise with the training means and standard deviations.
mean = X_train.mean(axis=0)
std = X_train.std(axis=0)
Z_train = (X_train - mean) / std
Z_test = (X_test - mean) / std

# Least squares: a column of ones, then solve (A^T A) w = A^T y.
A_train = np.hstack([np.ones((len(Z_train), 1)), Z_train])
A_test = np.hstack([np.ones((len(Z_test), 1)), Z_test])
w = np.linalg.solve(A_train.T @ A_train, A_train.T @ y_train)
linear_rmse = np.sqrt(np.mean((y_test - A_test @ w) ** 2))
print("Linear RMSE:", round(linear_rmse, 2))
```

Output for Task 1:

```
Train rows: 483  Test rows: 122
Baseline RMSE: 36783.75
Linear RMSE: 18318.81
```

The RMSE (root mean squared error) is the typical size of a prediction error, in the units of the target. A smaller RMSE is a better prediction. The seed fixes which rows land in the test part, so every member of the group gets the same split.

A classification task predicts the column `flag`, which you add in Part B. The baseline and the measure change. Here is the code filled in for Task 3:

```
import numpy as np

# clean is your cleaned and merged table from Part B,
# with the column flag (1 for "Fail", 0 otherwise).
features = [
    "attendance_pct", "study_hours_week", "previous_gpa",
    "late_assignments", "transport_cost", "class_size",
]
X = clean[features].values
y = clean["flag"].values

rng = np.random.default_rng(2003)
in_train = rng.random(len(clean)) < 0.8
X_train, X_test = X[in_train], X[~in_train]
y_train, y_test = y[in_train], y[~in_train]
print("Train rows:", len(y_train), " Test rows:", len(y_test))

# Baseline: predict the more common training label for every test row.
if y_train.mean() > 0.5:
    majority = 1
else:
    majority = 0
baseline_accuracy = np.mean(y_test == majority)
print("Baseline accuracy:", round(baseline_accuracy, 3))

# Standardise with the training means and standard deviations.
mean = X_train.mean(axis=0)
std = X_train.std(axis=0)
Z_train = (X_train - mean) / std
Z_test = (X_test - mean) / std

# Least squares on the 0/1 flag, then call a score of 0.5 or more a 1.
A_train = np.hstack([np.ones((len(Z_train), 1)), Z_train])
A_test = np.hstack([np.ones((len(Z_test), 1)), Z_test])
w = np.linalg.solve(A_train.T @ A_train, A_train.T @ y_train)
predicted = np.where(A_test @ w >= 0.5, 1, 0)
linear_accuracy = np.mean(predicted == y_test)
print("Linear accuracy:", round(linear_accuracy, 3))
```

Output for Task 3:

```
Train rows: 487  Test rows: 137
Baseline accuracy: 0.752
Linear accuracy: 0.788
```

The accuracy is the share of test rows predicted correctly. The baseline always predicts the more common label. A model that cannot beat it has learned nothing from the features.

What to Hand In

Hand in one zip file named `groupNN_phase1.zip` that holds:

- your program `groupNN_phase1.py` or notebook `groupNN_phase1.ipynb`, which must run from start to end on the two data files;
- `groupNN_clean.csv` and `groupNN_dashboard.png`;
- `groupNN_report.pdf`, with the roll numbers of all group members on the first page.

Every member of the group must be able to explain every line of the program. The deadline will be announced in class.

Part	What is marked	Marks
A	files read and every problem listed	4
B	cleaning steps in order, check values met	8
C	five answers, each with its number and meaning	6
D	arrays, z-scores, correlations, split and both baselines	8
E	seven charts, labelled, and the saved dashboard	8
F	clean file saved, and a clear report	6
	Total	40

Task 1: Apartment Rents in Karachi

Group 1 Target for the final project: `monthly_rent` (regression)

Saahil Property Advisers lists apartments for rent across Karachi. Each row is one listing from 2025. The office wants to suggest a fair rent when a new flat comes in. Some listings were typed in a hurry, and the export ran twice for a few of them. The data are invented for practice.

Files in the folder `Project_Phase1_Data`:

- data: `task01_karachi_rents.csv`, 632 rows
- lookup: `task01_karachi_rents_lookup.csv`, 8 rows

Columns of the data file

Column	Meaning	Unit or values
`listing_id`	label of the listing	text, one per row
`listed_on`	date the flat was listed	date, written YYYY-MM-DD
`area`	area of the city	Clifton, Defence, Gulshan Iqbal, North Nazimabad, Gulistan Jauhar, Saddar, Korangi, Malir
`furnished`	the flat comes with furniture	Yes, No
`size_sqft`	covered area	square feet
`bedrooms`	number of bedrooms	count
`floor`	floor of the flat (0 is ground)	floor
`building_age_years`	age of the building	years
`monthly_rent`	asking rent per month (target)	PKR, written as text such as Rs 45,000

Columns of the lookup file

Column	Meaning	Unit or values
`area`	area of the city	one row for each value
`distance_to_saddar_km`	road distance from Saddar	km
`zone`	part of the city	Central, East, South

First lines of the data file

```
listing_id,listed_on,area,furnished,size_sqft,bedrooms,floor,building_age_years,
    monthly_rent
KR0001,2025-01-01,North Nazimabad,No,980,2,4,,"Rs 72,000"
KR0002,2025-01-02,Gulshan Iqbal,No,1520,4,2,23,"Rs 96,000"
KR0003,2025-01-02,Korangi,No,890,1,2,6,"Rs 54,000"
```

Part B: Cleaning Rules for This Task

Apply these steps in this order. The check values below assume this order.

1. Drop rows that are exact copies of an earlier row.
2. Clean `area`: remove spaces at both ends, then write each word with a capital first letter.
3. Turn `monthly_rent` from text such as `Rs 45,000` into numbers.
4. Turn `listed_on` into dates.
5. Remove the rows where `monthly_rent` is missing.
6. Remove the rows where `size_sqft` is below 300 or above 5,000. Keep the rows where it is missing.

7. Fill the missing values of `floor` and of `building_age_years` with the median of that column.
8. Merge with the lookup file on `area`, keeping every row of your table.

Check values: your program should print these

Rows in the raw file	632
Rows after step 1 (exact copies dropped)	620
Rows after step 5 (missing target removed)	611
Rows after step 6 (impossible `size_sqft` removed)	605
Median used to fill `floor`	6.00
Median used to fill `building_age_years`	18.00
Rows and columns of the final table	605 and 11
Missing values left in the final table	0
Mean of `monthly_rent`	92,709.09
Part D: train rows and test rows (seed 2001)	483 and 122
Part D: baseline RMSE (predict the training mean)	36,783.75
Part D: least-squares RMSE	18,318.81

Part C: Questions for This Task

C1. For each `area`, give the number of listings and the mean `monthly_rent`. Which `area` has the highest value?

C2. Make a pivot table of the mean `monthly_rent`, with `area` down the side and `furnished` across the top. Which pair of `area` and `furnished` gives the highest value?

C3. Group the rows by the month of `listed_on`. Which month has the highest value of the mean `monthly_rent`, and which has the lowest?

C4. Compare the mean `monthly_rent` across the groups of `zone` from the lookup file. Then give the correlation between `distance_to_saddar_km` and `monthly_rent`, using `np.corrcoef`.

C5. What share of furnished listings in Clifton and Defence ask more than Rs 150,000 a month?

Parts D and E: Names for This Task

Features for Part D, in this order: `size_sqft`, `bedrooms`, `floor`, `building_age_years`, `distance_to_saddar_km`. The value to predict is `monthly_rent`. Use the seed 2001 for the split.

For Part E, show `monthly_rent` with a histogram coloured by `furnished`. Compare the groups of `area` with a box plot. For the outlier check (E7), use `monthly_rent`.

Next step. In the final project you will train regression models that predict `monthly_rent` for a new listing. They must beat the RMSE values above.

Task 2: Used Car Prices in Lahore

Group 2 Target for the final project: `price` (regression)

Ravi Auto Bazaar collects the adverts of six car showrooms in Lahore. Each row is one car offered for sale. Buyers want to know whether an asking price is fair. The showroom names were typed by different clerks. The data are invented for practice.

Files in the folder `Project_Phase1_Data`:

- data: `task02_lahore_used_cars.csv`, 714 rows
- lookup: `task02_lahore_used_cars_lookup.csv`, 6 rows

Columns of the data file

Column	Meaning	Unit or values
`ad_id`	label of the car	text, one per row
`advertised_on`	date of the advert	date, written YYYY-MM-DD
`showroom`	showroom that sells the car	Mall Road Autos, Ferozepur Road Motors, Gulberg Cars, Johar Town Motors, Shadman Autos, Raiwind Road Cars
`body_type`	shape of the car	Hatchback, Sedan, SUV
`age_years`	age of the car	years
`mileage_km`	distance driven	km
`engine_cc`	engine size	cc
`previous_owners`	number of earlier owners	count
`price`	asking price (target)	PKR, written as text such as Rs 45,000

Columns of the lookup file

Column	Meaning	Unit or values
`showroom`	showroom that sells the car	one row for each value
`showroom_rating`	average buyer rating of the showroom	1 to 5
`area_type`	kind of area around the showroom	Commercial, Residential, Suburban

First lines of the data file

```
ad_id,advertised_on,showroom,body_type,age_years,mileage_km,engine_cc,previous_owners,price
LC0001,2025-01-01,Mall Road Autos,Hatchback,3,39100,1000,1,"Rs 3,790,000"
LC0002,2025-01-01,Raiwind Road Cars,Hatchback,12,168900,1000,2,"Rs 1,660,000"
LC0003,2025-01-01,Raiwind Road Cars,Hatchback,2,47000,1200,4,"Rs 3,410,000"
```

Part B: Cleaning Rules for This Task

Apply these steps in this order. The check values below assume this order.

1. Drop rows that are exact copies of an earlier row.
2. Clean `showroom`: remove spaces at both ends, then write each word with a capital first letter.
3. Turn `price` from text such as `Rs 45,000` into numbers.
4. Turn `advertised_on` into dates.
5. Remove the rows where `price` is missing.
6. Remove the rows where `engine_cc` is below 600 or above 3,000. Keep the rows where it is missing.

7. Fill the missing values of `mileage_km` and of `previous_owners` with the median of that column.
8. Merge with the lookup file on `showroom`, keeping every row of your table.

Check values: your program should print these

Rows in the raw file	714
Rows after step 1 (exact copies dropped)	700
Rows after step 5 (missing target removed)	690
Rows after step 6 (impossible `engine_cc` removed)	684
Median used to fill `mileage_km`	111,300.00
Median used to fill `previous_owners`	3.00
Rows and columns of the final table	684 and 11
Missing values left in the final table	0
Mean of `price`	3,240,614.04
Part D: train rows and test rows (seed 2002)	540 and 144
Part D: baseline RMSE (predict the training mean)	1,538,950.71
Part D: least-squares RMSE	519,698.90

Part C: Questions for This Task

C1. For each `showroom`, give the number of cars and the mean `price`. Which `showroom` has the highest value?

C2. Make a pivot table of the mean `price`, with `showroom` down the side and `body_type` across the top. Which pair of `showroom` and `body_type` gives the highest value?

C3. Group the rows by the month of `advertised_on`. Which month has the highest value of the mean `price`, and which has the lowest?

C4. Compare the mean `price` across the groups of `area_type` from the lookup file. Then give the correlation between `showroom_rating` and `price`, using `np.corrcoef`.

C5. How much lower is the mean price of sedans driven more than 100,000 km than the mean price of sedans driven less than 50,000 km?

Parts D and E: Names for This Task

Features for Part D, in this order: `age_years`, `mileage_km`, `engine_cc`, `previous_owners`, `showroom_rating`. The value to predict is `price`. Use the seed 2002 for the split.

For Part E, show `price` with a histogram coloured by `body_type`. Compare the groups of `showroom` with a box plot. For the outlier check (E7), use `price`.

Next step. In the final project you will train regression models that predict `price` for a new car. They must beat the RMSE values above.

Task 3: Final Results at a Multan College

Group 3 Target for the final project: `result` (classification)

Chenab Degree College in Multan keeps a record for every second-year student. Each row is one student. The college wants to find students at risk of failing early in the year. The department names come from six different offices. The data are invented for practice.

Files in the folder `Project_Phase1_Data`:

- data: `task03_multan_results.csv`, 653 rows
- lookup: `task03_multan_results_lookup.csv`, 6 rows

Columns of the data file

Column	Meaning	Unit or values
`student_id`	label of the student	text, one per row
`enrolled_on`	date of enrolment	date, written YYYY-MM-DD
`department`	department of the student	Mathematics, Physics, Chemistry, Botany, Economics, Statistics
`hostel`	the student lives in the hostel	Yes, No
`attendance_pct`	share of lectures attended	percent
`study_hours_week`	hours of self-study per week	hours
`previous_gpa`	grade point average of first year	0 to 4
`late_assignments`	assignments handed in late	count
`transport_cost`	monthly cost of travel to college	PKR, written as text such as Rs 45,000
`result`	final result (target)	Fail or Pass

Columns of the lookup file

Column	Meaning	Unit or values
`department`	department of the student	one row for each value
`class_size`	students in one lecture room	count
`faculty`	faculty of the department	Life, Mathematical, Physical, Social

First lines of the data file

```
student_id,enrolled_on,department,hostel,attendance_pct,study_hours_week,previous_gpa,
    late_assignments,transport_cost,result
MC0001,2025-01-01,Economics,No,74,9.2,3.35,3,"Rs 4,100",Pass
MC0002,2025-01-01,Statistics,No,81,7.8,,0,Rs 0,Pass
MC0003,2025-01-02,Botany,No,85,11.2,3.03,0,"Rs 2,400",Pass
```

Part B: Cleaning Rules for This Task

Apply these steps in this order. The check values below assume this order.

1. Drop rows that are exact copies of an earlier row.
2. Clean `department`: remove spaces at both ends, then write each word with a capital first letter.
3. Turn `transport_cost` from text such as `Rs 45,000` into numbers.
4. Turn `enrolled_on` into dates.
5. Remove the rows where `result` is missing.

6. Remove the rows where `attendance_pct` is below 0 or above 100. Keep the rows where it is missing.
7. Fill the missing values of `study_hours_week` and of `previous_gpa` with the median of that column.
8. Merge with the lookup file on `department`, keeping every row of your table.
9. Add a column `flag` that is 1 where `result` is `Fail` and 0 otherwise.

Check values: your program should print these

Rows in the raw file	653
Rows after step 1 (exact copies dropped)	640
Rows after step 5 (missing target removed)	630
Rows after step 6 (impossible `attendance_pct` removed)	624
Median used to fill `study_hours_week`	9.10
Median used to fill `previous_gpa`	2.86
Rows and columns of the final table	624 and 13
Missing values left in the final table	0
Share of rows with `flag` equal to 1	0.271
Part D: train rows and test rows (seed 2003)	487 and 137
Part D: baseline accuracy (always the more common label)	0.752
Part D: least-squares accuracy	0.788

Part C: Questions for This Task

C1. For each `department`, give the number of students and the share of `result` equal to `Fail` (the mean of `flag`). Which `department` has the highest value?

C2. Make a pivot table of the share of `result` equal to `Fail` (the mean of `flag`), with `department` down the side and `hostel` across the top. Which pair of `department` and `hostel` gives the highest value?

C3. Group the rows by the month of `enrolled_on`. Which month has the highest value of the share of `result` equal to `Fail` (the mean of `flag`), and which has the lowest?

C4. Compare the share of `result` equal to `Fail` (the mean of `flag`) across the groups of `faculty` from the lookup file. Then give the correlation between `class_size` and `flag`, using `np.corrcoef`.

C5. Among students with attendance below 60 percent, what share failed?

Parts D and E: Names for This Task

Features for Part D, in this order: `attendance_pct`, `study_hours_week`, `previous_gpa`, `late_assignments`, `transport_cost`, `class_size`. The value to predict is `flag`. Use the seed 2003 for the split.

For Part E, show `result` with a count plot coloured by `hostel`. Compare the groups of `department` with a bar plot of the mean of `flag`. For the outlier check (E7), use your strongest feature from step D3.

Next step. In the final project you will train classification models that predict whether `result` is `Fail` for a new student. They must beat the accuracies above.

Task 4: Wheat Yield on Punjab Farms

Group 4 Target for the final project: `yield_maunds_per_acre` (regression)

Kisan Advisory Service visits wheat farms in seven districts of southern Punjab. Each row is one farm in the 2024 to 2025 season. The service wants to tell a farmer, before harvest, what yield to expect. Field staff wrote the district names in their own way. The data are invented for practice.

Files in the folder `Project_Phase1_Data`:

- data: `task04_punjab_wheat_yield.csv`, 571 rows
- lookup: `task04_punjab_wheat_yield_lookup.csv`, 7 rows

Columns of the data file

Column	Meaning	Unit or values
`farm_id`	label of the farm	text, one per row
`sown_on`	date the wheat was sown	date, written YYYY-MM-DD
`district`	district of the farm	Bahawalnagar, Vehari, Sahiwal, Okara, Khanewal, Pakpattan, Lodhran
`seed_type`	kind of seed	Certified, Saved
`fertiliser_bags_per_acre`	bags of fertiliser per acre	bags
`sowing_delay_days`	days sown after 1 November	days
`irrigations`	number of times the field was watered	count
`farm_size_acres`	size of the farm	acres
`input_cost_per_acre`	money spent on inputs per acre	PKR, written as text such as Rs 45,000
`yield_maunds_per_acre`	wheat harvested per acre (target)	maunds

Columns of the lookup file

Column	Meaning	Unit or values
`district`	district of the farm	one row for each value
`canal_water_days`	days of canal water in the season	days
`division`	division of the district	Bahawalpur, Multan, Sahiwal

First lines of the data file

```
farm_id,sown_on,district,seed_type,fertiliser_bags_per_acre,sowing_delay_days,irrigations,
    farm_size_acres,input_cost_per_acre,yield_maunds_per_acre
WF0001,2024-10-20,Sahiwal,Saved,4.2,0,4,5.7,"Rs 50,200",50.3
WF0002,2024-10-20,Lodhran,Certified,4.8,0,3,4.6,"Rs 60,300",39.7
WF0003,2024-10-20,Vehari,Saved,2.1,0,5,11.8,"Rs 45,400",37.6
```

Part B: Cleaning Rules for This Task

Apply these steps in this order. The check values below assume this order.

1. Drop rows that are exact copies of an earlier row.
2. Clean `district`: remove spaces at both ends, then write each word with a capital first letter.
3. Turn `input_cost_per_acre` from text such as `Rs 45,000` into numbers.
4. Turn `sown_on` into dates.
5. Remove the rows where `yield_maunds_per_acre` is missing.
6. Remove the rows where `fertiliser_bags_per_acre` is below 0 or above 8. Keep the rows where it is missing.

7. Fill the missing values of `irrigations` and of `farm_size_acres` with the median of that column.
8. Merge with the lookup file on `district`, keeping every row of your table.

Check values: your program should print these

Rows in the raw file	571
Rows after step 1 (exact copies dropped)	560
Rows after step 5 (missing target removed)	552
Rows after step 6 (impossible `fertiliser_bags_per_acre` removed)	546
Median used to fill `irrigations`	4.00
Median used to fill `farm_size_acres`	12.50
Rows and columns of the final table	546 and 12
Missing values left in the final table	0
Mean of `yield_maunds_per_acre`	39.04
Part D: train rows and test rows (seed 2004)	456 and 90
Part D: baseline RMSE (predict the training mean)	5.69
Part D: least-squares RMSE	3.34

Part C: Questions for This Task

C1. For each `district`, give the number of farms and the mean `yield_maunds_per_acre`. Which `district` has the highest value?

C2. Make a pivot table of the mean `yield_maunds_per_acre`, with `district` down the side and `seed_type` across the top. Which pair of `district` and `seed_type` gives the highest value?

C3. Group the rows by the month of `sown_on`. Which month has the highest value of the mean `yield_maunds_per_acre`, and which has the lowest?

C4. Compare the mean `yield_maunds_per_acre` across the groups of `division` from the lookup file. Then give the correlation between `canal_water_days` and `yield_maunds_per_acre`, using `np.corrcoef`.

C5. Within Sahiwal district, how many more maunds per acre does certified seed give, on average, than saved seed?

Parts D and E: Names for This Task

Features for Part D, in this order: `fertiliser_bags_per_acre`, `sowing_delay_days`, `irrigations`, `farm_size_acres`, `input_cost_per_acre`, `canal_water_days`. The value to predict is `yield_maunds_per_acre`. Use the seed 2004 for the split.

For Part E, show `yield_maunds_per_acre` with a histogram coloured by `seed_type`. Compare the groups of `district` with a box plot. For the outlier check (E7), use `yield_maunds_per_acre`.

Next step. In the final project you will train regression models that predict `yield_maunds_per_acre` for a new farm. They must beat the RMSE values above.

Task 5: Customers Leaving a Telecom Company in Islamabad

Group 5 Target for the final project: `churned` (classification)

Margalla Telecom sells mobile connections in seven areas of Islamabad. Each row is one customer. The company wants to know which customers are likely to leave, so it can call them first. The area names came from shop forms filled in by hand. The data are invented for practice.

Files in the folder `Project_Phase1_Data`:

- data: `task05_islamabad_churn.csv`, 796 rows
- lookup: `task05_islamabad_churn_lookup.csv`, 7 rows

Columns of the data file

Column	Meaning	Unit or values
`customer_id`	label of the customer	text, one per row
`last_recharge_on`	date of the last recharge or bill	date, written YYYY-MM-DD
`area`	area where the customer lives	Blue Area, Bahria Town, Bara Kahu, Tarnol, Saddar, Golra, Shahzad Town
`plan`	how the customer pays	Prepaid, Postpaid
`tenure_months`	months with the company	months
`data_gb_month`	mobile data used per month	GB
`complaints_6m`	complaints in the last six months	count
`dropped_calls_week`	calls dropped per week	count
`monthly_bill`	average amount paid per month	PKR, written as text such as Rs 45,000
`churned`	the customer left in the next three months (target)	Yes or No

Columns of the lookup file

Column	Meaning	Unit or values
`area`	area where the customer lives	one row for each value
`towers`	mobile towers in the area	count
`area_type`	kind of area	Rural, Urban

First lines of the data file

```
customer_id,last_recharge_on,area,plan,tenure_months,data_gb_month,complaints_6m,
    dropped_calls_week,monthly_bill,churned
MT0001,2025-01-01,Tarnol,Postpaid,29,2.6,1,1,"Rs 1,910",No
MT0002,2025-01-02,Tarnol,Postpaid,14,93.3,0,6,"Rs 1,760",No
MT0003,2025-01-03,Golra,Postpaid,24,6.1,2,3,Rs 800,Yes
```

Part B: Cleaning Rules for This Task

Apply these steps in this order. The check values below assume this order.

1. Drop rows that are exact copies of an earlier row.
2. Clean `area`: remove spaces at both ends, then write each word with a capital first letter.
3. Turn `monthly_bill` from text such as `Rs 45,000` into numbers.
4. Turn `last_recharge_on` into dates.
5. Remove the rows where `churned` is missing.
6. Remove the rows where `tenure_months` is below 0 or above 120. Keep the rows where it is missing.

7. Fill the missing values of `data_gb_month` and of `dropped_calls_week` with the median of that column.
8. Merge with the lookup file on `area`, keeping every row of your table.
9. Add a column `flag` that is 1 where `churned` is `Yes` and 0 otherwise.

Check values: your program should print these

Rows in the raw file	796
Rows after step 1 (exact copies dropped)	780
Rows after step 5 (missing target removed)	768
Rows after step 6 (impossible `tenure_months` removed)	760
Median used to fill `data_gb_month`	12.10
Median used to fill `dropped_calls_week`	3.00
Rows and columns of the final table	760 and 13
Missing values left in the final table	0
Share of rows with `flag` equal to 1	0.286
Part D: train rows and test rows (seed 2005)	604 and 156
Part D: baseline accuracy (always the more common label)	0.737
Part D: least-squares accuracy	0.846

Part C: Questions for This Task

C1. For each `area`, give the number of customers and the share of `churned` equal to `Yes` (the mean of `flag`). Which `area` has the highest value?

C2. Make a pivot table of the share of `churned` equal to `Yes` (the mean of `flag`), with `area` down the side and `plan` across the top. Which pair of `area` and `plan` gives the highest value?

C3. Group the rows by the month of `last_recharge_on`. Which month has the highest value of the share of `churned` equal to `Yes` (the mean of `flag`), and which has the lowest?

C4. Compare the share of `churned` equal to `Yes` (the mean of `flag`) across the groups of `area_type` from the lookup file. Then give the correlation between `towers` and `flag`, using `np.corrcoef`.

C5. What share of prepaid customers with three or more complaints left?

Parts D and E: Names for This Task

Features for Part D, in this order: `tenure_months`, `data_gb_month`, `complaints_6m`, `dropped_calls_week`, `monthly_bill`, `towers`. The value to predict is `flag`. Use the seed 2005 for the split.

For Part E, show `churned` with a count plot coloured by `plan`. Compare the groups of `area` with a bar plot of the mean of `flag`. For the outlier check (E7), use your strongest feature from step D3.

Next step. In the final project you will train classification models that predict whether `churned` is `Yes` for a new customer. They must beat the accuracies above.

Task 6: Small Loans in Faisalabad

Group 6 Target for the final project: `defaulted` (classification)

Lyallpur Microfinance gives small business loans from six branches in and around Faisalabad. Each row is one loan made in 2025. The lender wants to spot loans that may not be repaid before it approves them. Branch names were entered by loan officers on tablets. The data are invented for practice.

Files in the folder `Project_Phase1_Data`:

- data: `task06_faisalabad_microloans.csv`, 663 rows
- lookup: `task06_faisalabad_microloans_lookup.csv`, 6 rows

Columns of the data file

Column	Meaning	Unit or values
`loan_id`	label of the loan	text, one per row
`disbursed_on`	date the money was paid out	date, written YYYY-MM-DD
`branch`	branch that made the loan	Jhang Road, Samanabad, Madina Town, Ghulam Muhammadabad, Peoples Colony, Jaranwala
`business_type`	business the loan pays for	Grocery Shop, Tailoring, Livestock, Rickshaw, Vegetable Cart
`monthly_income`	income of the household per month	PKR (number)
`household_size`	people in the household	count
`previous_loans_repaid`	earlier loans repaid in full	count
`business_age_years`	age of the business	years
`loan_amount`	amount lent	PKR, written as text such as Rs 45,000
`defaulted`	the loan was not repaid on time (target)	Yes or No

Columns of the lookup file

Column	Meaning	Unit or values
`branch`	branch that made the loan	one row for each value
`loan_officers`	loan officers at the branch	count
`setting`	where the branch is	Inner City, Outer Town, Suburb

First lines of the data file

```
loan_id,disbursed_on,branch,business_type,monthly_income,household_size,
    previous_loans_repaid,business_age_years,loan_amount,defaulted
LM0001,2025-01-01,Madina Town,Tailoring,68100,4,0,18,"Rs 49,000",No
LM0002,2025-01-02,Peoples Colony,Grocery Shop,24400,6,1,20,"Rs 38,000",Yes
LM0003,2025-01-02,Peoples Colony,Grocery Shop,66900,10,4,16,"Rs 32,000",No
```

Part B: Cleaning Rules for This Task

Apply these steps in this order. The check values below assume this order.

1. Drop rows that are exact copies of an earlier row.
2. Clean `branch`: remove spaces at both ends, then write each word with a capital first letter.
3. Turn `loan_amount` from text such as `Rs 45,000` into numbers.

4. Turn `disbursed_on` into dates.
5. Remove the rows where `defaulted` is missing.
6. Remove the rows where `household_size` is below 1 or above 15. Keep the rows where it is missing.
7. Fill the missing values of `monthly_income` and of `business_age_years` with the median of that column.
8. Merge with the lookup file on `branch`, keeping every row of your table.
9. Add a column `flag` that is 1 where `defaulted` is `Yes` and 0 otherwise.

Check values: your program should print these

Rows in the raw file	663
Rows after step 1 (exact copies dropped)	650
Rows after step 5 (missing target removed)	640
Rows after step 6 (impossible `household_size` removed)	635
Median used to fill `monthly_income`	51,900.00
Median used to fill `business_age_years`	10.00
Rows and columns of the final table	635 and 13
Missing values left in the final table	0
Share of rows with `flag` equal to 1	0.261
Part D: train rows and test rows (seed 2006)	504 and 131
Part D: baseline accuracy (always the more common label)	0.756
Part D: least-squares accuracy	0.817

Part C: Questions for This Task

C1. For each `branch`, give the number of loans and the share of `defaulted` equal to `Yes` (the mean of `flag`). Which `branch` has the highest value?

C2. Make a pivot table of the share of `defaulted` equal to `Yes` (the mean of `flag`), with `branch` down the side and `business_type` across the top. Which pair of `branch` and `business_type` gives the highest value?

C3. Group the rows by the month of `disbursed_on`. Which month has the highest value of the share of `defaulted` equal to `Yes` (the mean of `flag`), and which has the lowest?

C4. Compare the share of `defaulted` equal to `Yes` (the mean of `flag`) across the groups of `setting` from the lookup file. Then give the correlation between `loan_officers` and `flag`, using `np.corrcoef`.

C5. Among loans larger than Rs 100,000, what share were not repaid?

Parts D and E: Names for This Task

Features for Part D, in this order: `monthly_income`, `household_size`, `previous_loans_repaid`, `business_age_years`, `loan_amount`, `loan_officers`. The value to predict is `flag`. Use the seed 2006 for the split.

For Part E, show `defaulted` with a count plot coloured by `business_type`. Compare the groups of `branch` with a bar plot of the mean of `flag`. For the outlier check (E7), use your strongest feature from step D3.

Next step. In the final project you will train classification models that predict whether `defaulted` is `Yes` for a new loan. They must beat the accuracies above.

Task 7: Food Delivery Times in Peshawar

Group 7 Target for the final project: `delivery_minutes` (regression)

Qissa Khwani Eats delivers restaurant food across Peshawar. Each row is one delivered order. The app wants to show customers a delivery time they can trust. Riders type the pickup area into their phones. The data are invented for practice.

Files in the folder `Project_Phase1_Data`:

- data: `task07_peshawar_deliveries.csv`, 734 rows
- lookup: `task07_peshawar_deliveries_lookup.csv`, 7 rows

Columns of the data file

Column	Meaning	Unit or values
`order_id`	label of the order	text, one per row
`ordered_on`	date of the order	date, written YYYY-MM-DD
`pickup_area`	area of the restaurant	Hayatabad, University Town, Saddar, Gulbahar, Board Bazaar, Warsak Road, Ring Road
`time_slot`	time of day of the order	Lunch, Evening, Late Night
`distance_km`	distance from restaurant to customer	km
`items`	dishes in the order	count
`rider_experience_months`	months the rider has worked	months
`rain_mm`	rain in the hour of the order	mm
`order_value`	bill of the order	PKR, written as text such as Rs 45,000
`delivery_minutes`	minutes from order to door (target)	minutes

Columns of the lookup file

Column	Meaning	Unit or values
`pickup_area`	area of the restaurant	one row for each value
`traffic_index`	usual traffic in the area	1 (light) to 10 (heavy)
`zone`	part of the city	Central, North, South, West

First lines of the data file

```
order_id,ordered_on,pickup_area,time_slot,distance_km,items,rider_experience_months,rain_mm
    ,order_value,delivery_minutes
PD0001,2025-01-01,Board Bazaar  ,Evening,6.1,6,,0.0,Rs 450,38
PD0002,2025-01-04,Warsak Road,Evening,3.8,6,28,8.0,"Rs 1,240",46
PD0003,2025-01-04,Ring Road,Late Night,5.3,2,40,0.0,"Rs 1,200",21
```

Part B: Cleaning Rules for This Task

Apply these steps in this order. The check values below assume this order.

1. Drop rows that are exact copies of an earlier row.
2. Clean `pickup_area`: remove spaces at both ends, then write each word with a capital first letter.
3. Turn `order_value` from text such as `Rs 45,000` into numbers.
4. Turn `ordered_on` into dates.
5. Remove the rows where `delivery_minutes` is missing.

6. Remove the rows where `distance_km` is below 0.1 or above 40. Keep the rows where it is missing.
7. Fill the missing values of `rider_experience_months` and of `rain_mm` with the median of that column.
8. Merge with the lookup file on `pickup_area`, keeping every row of your table.

Check values: your program should print these

Rows in the raw file	734
Rows after step 1 (exact copies dropped)	720
Rows after step 5 (missing target removed)	709
Rows after step 6 (impossible `distance_km` removed)	703
Median used to fill `rider_experience_months`	24.00
Median used to fill `rain_mm`	0.00
Rows and columns of the final table	703 and 12
Missing values left in the final table	0
Mean of `delivery_minutes`	37.29
Part D: train rows and test rows (seed 2007)	580 and 123
Part D: baseline RMSE (predict the training mean)	11.79
Part D: least-squares RMSE	6.35

Part C: Questions for This Task

C1. For each `pickup_area`, give the number of orders and the mean `delivery_minutes`. Which `pickup_area` has the highest value?

C2. Make a pivot table of the mean `delivery_minutes`, with `pickup_area` down the side and `time_slot` across the top. Which pair of `pickup_area` and `time_slot` gives the highest value?

C3. Group the rows by the month of `ordered_on`. Which month has the highest value of the mean `delivery_minutes`, and which has the lowest?

C4. Compare the mean `delivery_minutes` across the groups of `zone` from the lookup file. Then give the correlation between `traffic_index` and `delivery_minutes`, using `np.corrcoef`.

C5. For evening orders with some rain (more than 0 mm), what is the median delivery time?

Parts D and E: Names for This Task

Features for Part D, in this order: `distance_km`, `items`, `rider_experience_months`, `rain_mm`, `order_value`, `traffic_index`. The value to predict is `delivery_minutes`. Use the seed 2007 for the split.

For Part E, show `delivery_minutes` with a histogram coloured by `time_slot`. Compare the groups of `pickup_area` with a box plot. For the outlier check (E7), use `delivery_minutes`.

Next step. In the final project you will train regression models that predict `delivery_minutes` for a new order. They must beat the RMSE values above.

Task 8: Diabetes Screening Camps near Hyderabad

Group 8 Target for the final project: `diabetic` (classification)

Indus Health Screening Camps tests adults for diabetes in six towns around Hyderabad. Each row is one adult screened in 2025. The camps want to invite people at higher risk for a full test first. Volunteers wrote the town names on paper forms. The data are invented for practice.

Files in the folder `Project_Phase1_Data`:

- data: `task08_hyderabad_screening.csv`, 612 rows
- lookup: `task08_hyderabad_screening_lookup.csv`, 6 rows

Columns of the data file

Column	Meaning	Unit or values
`screening_id`	label of the person	text, one per row
`screened_on`	date of the screening	date, written YYYY-MM-DD
`town`	town of the camp	Latifabad, Qasimabad, Kotri, Tando Jam, Hala, Hyderabad Saddar
`family_history`	a parent or sibling has diabetes	Yes, No
`age`	age of the person	years
`bmi`	body mass index	kg per square metre
`waist_cm`	waist measurement	cm
`daily_steps`	steps walked per day	steps
`monthly_food_spend`	money spent on food per month	PKR, written as text such as Rs 45,000
`diabetic`	the full test confirmed diabetes (target)	Yes or No

Columns of the lookup file

Column	Meaning	Unit or values
`town`	town of the camp	one row for each value
`fast_food_outlets`	fast food outlets in the town	count
`setting`	kind of town	City, Town

First lines of the data file

```
screening_id,screened_on,town,family_history,age,bmi,waist_cm,daily_steps,
    monthly_food_spend,diabetic
HS0001,2025-01-01,Hala,No,33,22.6,94,6700,"Rs 12,800",No
HS0002,2025-01-01,kotri,No,49,28.2,97,6300,"Rs 14,300",No
HS0003,2025-01-03,Qasimabad,Yes,60,16.1,73,5600,"Rs 13,200",No
```

Part B: Cleaning Rules for This Task

Apply these steps in this order. The check values below assume this order.

1. Drop rows that are exact copies of an earlier row.
2. Clean `town`: remove spaces at both ends, then write each word with a capital first letter.
3. Turn `monthly_food_spend` from text such as `Rs 45,000` into numbers.
4. Turn `screened_on` into dates.
5. Remove the rows where `diabetic` is missing.
6. Remove the rows where `bmi` is below 12 or above 60. Keep the rows where it is missing.

7. Fill the missing values of `waist_cm` and of `daily_steps` with the median of that column.
8. Merge with the lookup file on `town`, keeping every row of your table.
9. Add a column `flag` that is 1 where `diabetic` is `Yes` and 0 otherwise.

Check values: your program should print these

Rows in the raw file	612
Rows after step 1 (exact copies dropped)	600
Rows after step 5 (missing target removed)	591
Rows after step 6 (impossible `bmi` removed)	585
Median used to fill `waist_cm`	97.00
Median used to fill `daily_steps`	5,500.00
Rows and columns of the final table	585 and 13
Missing values left in the final table	0
Share of rows with `flag` equal to 1	0.354
Part D: train rows and test rows (seed 2008)	460 and 125
Part D: baseline accuracy (always the more common label)	0.656
Part D: least-squares accuracy	0.760

Part C: Questions for This Task

C1. For each `town`, give the number of people and the share of `diabetic` equal to `Yes` (the mean of `flag`). Which `town` has the highest value?

C2. Make a pivot table of the share of `diabetic` equal to `Yes` (the mean of `flag`), with `town` down the side and `family_history` across the top. Which pair of `town` and `family_history` gives the highest value?

C3. Group the rows by the month of `screened_on`. Which month has the highest value of the share of `diabetic` equal to `Yes` (the mean of `flag`), and which has the lowest?

C4. Compare the share of `diabetic` equal to `Yes` (the mean of `flag`) across the groups of `setting` from the lookup file. Then give the correlation between `fast_food_outlets` and `flag`, using `np.corrcoef`.

C5. Among people aged 50 or more with a family history, what share were diabetic?

Parts D and E: Names for This Task

Features for Part D, in this order: `age`, `bmi`, `waist_cm`, `daily_steps`, `monthly_food_spend`, `fast_food_outlets`. The value to predict is `flag`. Use the seed 2008 for the split.

For Part E, show `diabetic` with a count plot coloured by `family_history`. Compare the groups of `town` with a bar plot of the mean of `flag`. For the outlier check (E7), use your strongest feature from step D3.

Next step. In the final project you will train classification models that predict whether `diabetic` is `Yes` for a new person. They must beat the accuracies above.

Task 9: Export Grade of Kinnow from Sargodha

Group 9 Target for the final project: `export_grade` (classification)

Sargodha Citrus Packers buys kinnow from orchards in six areas and packs them for export. Each row is one sample of fruit checked at the gate. The packer wants to predict, before sorting, which loads will pass the export check. Gate staff typed the orchard area in different ways. The data are invented for practice.

Files in the folder `Project_Phase1_Data`:

- data: `task09_sargodha_kinnow.csv`, 694 rows
- lookup: `task09_sargodha_kinnow_lookup.csv`, 6 rows

Columns of the data file

Column	Meaning	Unit or values
`sample_id`	label of the sample	text, one per row
`picked_on`	date the fruit was picked	date, written YYYY-MM-DD
`orchard_area`	area of the orchard	Bhalwal, Kot Momin, Sillanwali, Shahpur, Sargodha City, Bhera
`picking`	how the fruit was picked	Hand, Clipper
`fruit_weight_g`	mean weight of one fruit	grams
`brix`	sugar content of the juice	degrees Brix
`peel_mm`	thickness of the peel	mm
`days_in_store`	days between picking and checking	days
`price_per_crate`	price paid to the grower per crate	PKR, written as text such as Rs 45,000
`export_grade`	the sample passed the export check (target)	Yes or No

Columns of the lookup file

Column	Meaning	Unit or values
`orchard_area`	area of the orchard	one row for each value
`avg_tree_age_years`	average age of the trees	years
`soil`	main soil of the area	Clay Loam, Loam, Sandy Loam

First lines of the data file

```
sample_id,picked_on,orchard_area,picking,fruit_weight_g,brix,peel_mm,days_in_store,
    price_per_crate,export_grade
SK0001,2024-12-01,Shahpur,Hand,199,12.5,3.1,1,"Rs 2,010",Yes
SK0002,2024-12-01,Kot Momin,Hand,206,9.5,3.7,10,"Rs 2,430",No
SK0003,2024-12-01,BHERA,Hand,202,11.1,2.2,7,"Rs 2,140",Yes
```

Part B: Cleaning Rules for This Task

Apply these steps in this order. The check values below assume this order.

1. Drop rows that are exact copies of an earlier row.
2. Clean `orchard_area`: remove spaces at both ends, then write each word with a capital first letter.
3. Turn `price_per_crate` from text such as `Rs 45,000` into numbers.
4. Turn `picked_on` into dates.

5. Remove the rows where `export_grade` is missing.
6. Remove the rows where `fruit_weight_g` is below 50 or above 400. Keep the rows where it is missing.
7. Fill the missing values of `peel_mm` and of `days_in_store` with the median of that column.
8. Merge with the lookup file on `orchard_area`, keeping every row of your table.
9. Add a column `flag` that is 1 where `export_grade` is `Yes` and 0 otherwise.

Check values: your program should print these

Rows in the raw file	694
Rows after step 1 (exact copies dropped)	680
Rows after step 5 (missing target removed)	670
Rows after step 6 (impossible `fruit_weight_g` removed)	663
Median used to fill `peel_mm`	3.20
Median used to fill `days_in_store`	7.00
Rows and columns of the final table	663 and 13
Missing values left in the final table	0
Share of rows with `flag` equal to 1	0.425
Part D: train rows and test rows (seed 2009)	529 and 134
Part D: baseline accuracy (always the more common label)	0.567
Part D: least-squares accuracy	0.761

Part C: Questions for This Task

C1. For each `orchard_area`, give the number of samples and the share of `export_grade` equal to `Yes` (the mean of `flag`). Which `orchard_area` has the highest value?

C2. Make a pivot table of the share of `export_grade` equal to `Yes` (the mean of `flag`), with `orchard_area` down the side and `picking` across the top. Which pair of `orchard_area` and `picking` gives the highest value?

C3. Group the rows by the month of `picked_on`. Which month has the highest value of the share of `export_grade` equal to `Yes` (the mean of `flag`), and which has the lowest?

C4. Compare the share of `export_grade` equal to `Yes` (the mean of `flag`) across the groups of `soil` from the lookup file. Then give the correlation between `avg_tree_age_years` and `flag`, using `np.corrcoef`.

C5. Among samples with a Brix of 12 or more, what share reached export grade?

Parts D and E: Names for This Task

Features for Part D, in this order: `fruit_weight_g`, `brix`, `peel_mm`, `days_in_store`, `price_per_crate`, `avg_tree_age_years`. The value to predict is `flag`. Use the seed 2009 for the split.

For Part E, show `export_grade` with a count plot coloured by `picking`. Compare the groups of `orchard_area` with a bar plot of the mean of `flag`. For the outlier check (E7), use your strongest feature from step D3.

Next step. In the final project you will train classification models that predict whether `export_grade` is `Yes` for a new sample. They must beat the accuracies above.

Task 10: Unsafe Well Water in Quetta

Group 10 Target for the final project: `unsafe` (classification)

Bolan Water Testing Lab tests drinking water from wells around Quetta. Each row is one well tested in 2025. The lab wants to decide which wells to test first when its kits run short. The area names come from the request slips. The data are invented for practice.

Files in the folder `Project_Phase1_Data`:

- data: `task10_quetta_wells.csv`, 551 rows
- lookup: `task10_quetta_wells_lookup.csv`, 7 rows

Columns of the data file

Column	Meaning	Unit or values
`sample_id`	label of the well	text, one per row
`tested_on`	date of the test	date, written YYYY-MM-DD
`area`	area of the well	Kuchlak, Sariab, Hanna Urak, Nawa Killi, Satellite Town, Brewery Road, Samungli
`well_type`	kind of well	Tube Well, Dug Well, Hand Pump
`well_depth_m`	depth of the well	metres
`tds_ppm`	dissolved salts in the water	parts per million
`nitrate_mg_l`	nitrate in the water	mg per litre
`distance_to_latrine_m`	distance to the nearest latrine	metres
`drilling_cost`	cost of making the well	PKR, written as text such as Rs 45,000
`unsafe`	the water failed the safety test (target)	Yes or No

Columns of the lookup file

Column	Meaning	Unit or values
`area`	area of the well	one row for each value
`households_per_well`	households sharing one well	count
`terrain`	land around the area	Hill, Plain, Valley

First lines of the data file

```
sample_id,tested_on,area,well_type,well_depth_m,tds_ppm,nitrate_mg_l,distance_to_latrine_m,
    drilling_cost,unsafe
QW0001,2025-01-01,Sariab,Tube Well,77,810,20.1,9,"Rs 1,100,000",Yes
QW0002,2025-01-02,Hanna Urak,Tube Well,222,1160,33.0,35,"Rs 790,000",No
QW0003,2025-01-03,Brewery Road,Dug Well,85,800,60.0,81,"Rs 430,000",Yes
```

Part B: Cleaning Rules for This Task

Apply these steps in this order. The check values below assume this order.

1. Drop rows that are exact copies of an earlier row.
2. Clean `area`: remove spaces at both ends, then write each word with a capital first letter.
3. Turn `drilling_cost` from text such as `Rs 45,000` into numbers.
4. Turn `tested_on` into dates.
5. Remove the rows where `unsafe` is missing.

6. Remove the rows where `well_depth_m` is below 3 or above 600. Keep the rows where it is missing.
7. Fill the missing values of `nitrate_mg_l` and of `distance_to_latrine_m` with the median of that column.
8. Merge with the lookup file on `area`, keeping every row of your table.
9. Add a column `flag` that is 1 where `unsafe` is `Yes` and 0 otherwise.

Check values: your program should print these

Rows in the raw file	551
Rows after step 1 (exact copies dropped)	540
Rows after step 5 (missing target removed)	532
Rows after step 6 (impossible `well_depth_m` removed)	527
Median used to fill `nitrate_mg_l`	25.20
Median used to fill `distance_to_latrine_m`	24.00
Rows and columns of the final table	527 and 13
Missing values left in the final table	0
Share of rows with `flag` equal to 1	0.404
Part D: train rows and test rows (seed 2010)	434 and 93
Part D: baseline accuracy (always the more common label)	0.602
Part D: least-squares accuracy	0.710

Part C: Questions for This Task

C1. For each `area`, give the number of wells and the share of `unsafe` equal to `Yes` (the mean of `flag`). Which `area` has the highest value?

C2. Make a pivot table of the share of `unsafe` equal to `Yes` (the mean of `flag`), with `area` down the side and `well_type` across the top. Which pair of `area` and `well_type` gives the highest value?

C3. Group the rows by the month of `tested_on`. Which month has the highest value of the share of `unsafe` equal to `Yes` (the mean of `flag`), and which has the lowest?

C4. Compare the share of `unsafe` equal to `Yes` (the mean of `flag`) across the groups of `terrain` from the lookup file. Then give the correlation between `households_per_well` and `flag`, using `np.corrcoef`.

C5. Among dug wells closer than 15 m to a latrine, what share were unsafe?

Parts D and E: Names for This Task

Features for Part D, in this order: `well_depth_m`, `tds_ppm`, `nitrate_mg_l`, `distance_to_latrine_m`, `drilling_cost`, `households_per_well`. The value to predict is `flag`. Use the seed 2010 for the split.

For Part E, show `unsafe` with a count plot coloured by `well_type`. Compare the groups of `area` with a bar plot of the mean of `flag`. For the outlier check (E7), use your strongest feature from step D3.

Next step. In the final project you will train classification models that predict whether `unsafe` is `Yes` for a new well. They must beat the accuracies above.

Task 11: Faulty Footballs in Sialkot

Group 11 Target for the final project: `defective` (classification)

Sialkot Ball Works has footballs stitched in six workshops around the city. Each row is one ball checked by the quality team. The company wants to know which balls to check most closely. Workshop names were entered by the checkers. The data are invented for practice.

Files in the folder `Project_Phase1_Data`:

- data: `task11_sialkot_footballs.csv`, 775 rows
- lookup: `task11_sialkot_footballs_lookup.csv`, 6 rows

Columns of the data file

Column	Meaning	Unit or values
`ball_id`	label of the ball	text, one per row
`made_on`	date the ball was stitched	date, written YYYY-MM-DD
`workshop`	workshop that stitched the ball	Ugoki, Sambrial, Daska, Pasrur, Kotli Loharan, Sialkot Cantt
`shift`	work shift	Morning, Evening, Night
`panels`	panels in the ball	count
`stitch_tension`	tension setting of the thread	newtons
`humidity_pct`	humidity in the workshop	percent
`stitcher_experience_years`	experience of the stitcher	years
`material_cost`	cost of material for the ball	PKR, written as text such as Rs 45,000
`defective`	the ball failed the quality check (target)	Yes or No

Columns of the lookup file

Column	Meaning	Unit or values
`workshop`	workshop that stitched the ball	one row for each value
`stitchers`	stitchers at the workshop	count
`ownership`	who runs the workshop	Contract, Own

First lines of the data file

```
ball_id,made_on,workshop,shift,panels,stitch_tension,humidity_pct,stitcher_experience_years
    ,material_cost,defective
SB0001,2025-01-01, Kotli Loharan,Night,12,44.5,47,2.0,"Rs 1,180",Yes
SB0002,2025-01-01,Sambrial,Evening,12,51.2,74,4.5,"Rs 1,170",Yes
SB0003,2025-01-01,Pasrur,Evening,32,54.6,55,5.8,"Rs 1,340",Yes
```

Part B: Cleaning Rules for This Task

Apply these steps in this order. The check values below assume this order.

1. Drop rows that are exact copies of an earlier row.
2. Clean `workshop`: remove spaces at both ends, then write each word with a capital first letter.
3. Turn `material_cost` from text such as `Rs 45,000` into numbers.
4. Turn `made_on` into dates.
5. Remove the rows where `defective` is missing.
6. Remove the rows where `stitch_tension` is below 10 or above 100. Keep the rows where it is missing.

7. Fill the missing values of `humidity_pct` and of `stitcher_experience_years` with the median of that column.
8. Merge with the lookup file on `workshop`, keeping every row of your table.
9. Add a column `flag` that is 1 where `defective` is `Yes` and 0 otherwise.

Check values: your program should print these

Rows in the raw file	775
Rows after step 1 (exact copies dropped)	760
Rows after step 5 (missing target removed)	749
Rows after step 6 (impossible `stitch_tension` removed)	741
Median used to fill `humidity_pct`	61.00
Median used to fill `stitcher_experience_years`	5.10
Rows and columns of the final table	741 and 13
Missing values left in the final table	0
Share of rows with `flag` equal to 1	0.348
Part D: train rows and test rows (seed 2011)	606 and 135
Part D: baseline accuracy (always the more common label)	0.644
Part D: least-squares accuracy	0.674

Part C: Questions for This Task

C1. For each `workshop`, give the number of balls and the share of `defective` equal to `Yes` (the mean of `flag`). Which `workshop` has the highest value?

C2. Make a pivot table of the share of `defective` equal to `Yes` (the mean of `flag`), with `workshop` down the side and `shift` across the top. Which pair of `workshop` and `shift` gives the highest value?

C3. Group the rows by the month of `made_on`. Which month has the highest value of the share of `defective` equal to `Yes` (the mean of `flag`), and which has the lowest?

C4. Compare the share of `defective` equal to `Yes` (the mean of `flag`) across the groups of `ownership` from the lookup file. Then give the correlation between `stitchers` and `flag`, using `np.corrcoef`.

C5. On the night shift, what share of balls stitched by workers with less than 3 years of experience were faulty?

Parts D and E: Names for This Task

Features for Part D, in this order: `panels`, `stitch_tension`, `humidity_pct`, `stitcher_experience_years`, `material_cost`, `stitchers`. The value to predict is `flag`. Use the seed 2011 for the split.

For Part E, show `defective` with a count plot coloured by `shift`. Compare the groups of `workshop` with a bar plot of the mean of `flag`. For the outlier check (E7), use your strongest feature from step D3.

Next step. In the final project you will train classification models that predict whether `defective` is `Yes` for a new ball. They must beat the accuracies above.

Task 12: Returned Orders at an Online Clothing Store

Group 12 Target for the final project: `returned` (classification)

Anarkali Online Store sells clothes from Lahore to customers in eight cities. Each row is one order delivered in 2025. The store wants to know which orders are likely to come back, so it can check sizes before sending. Customers type their own city name at checkout. The data are invented for practice.

Files in the folder `Project_Phase1_Data`:

- data: `task12_online_clothing_returns.csv`, 836 rows
- lookup: `task12_online_clothing_returns_lookup.csv`, 8 rows

Columns of the data file

Column	Meaning	Unit or values
`order_id`	label of the order	text, one per row
`ordered_on`	date of the order	date, written YYYY-MM-DD
`city`	city of the customer	Lahore, Karachi, Islamabad, Rawalpindi, Multan, Faisalabad, Peshawar, Quetta
`category`	kind of item	Kurta, Shalwar Suit, Dupatta, Shoes
`items`	items in the order	count
`discount_pct`	discount given	percent
`previous_orders`	earlier orders by the same customer	count
`size_chart_views`	times the size chart was opened	count
`order_value`	bill of the order	PKR, written as text such as Rs 45,000
`returned`	the customer sent the order back (target)	Yes or No

Columns of the lookup file

Column	Meaning	Unit or values
`city`	city of the customer	one row for each value
`courier_days`	usual days for the courier to deliver	days
`province`	province or territory of the city	Balochistan, Federal, Khyber Pakhtunkhwa, Punjab, Sindh

First lines of the data file

```
order_id,ordered_on,city,category,items,discount_pct,previous_orders,size_chart_views,
    order_value,returned
A00001,2025-01-02,Karachi,Kurta,1,50,1,2,"Rs 6,940",Yes
A00002,2025-01-02,Faisalabad,Shalwar Suit,5,10,7,0,"Rs 5,750",No
A00003,2025-01-02,Peshawar,Kurta,5,10,2,0,"Rs 2,240",No
```

Part B: Cleaning Rules for This Task

Apply these steps in this order. The check values below assume this order.

1. Drop rows that are exact copies of an earlier row.
2. Clean `city`: remove spaces at both ends, then write each word with a capital first letter.
3. Turn `order_value` from text such as `Rs 45,000` into numbers.
4. Turn `ordered_on` into dates.

5. Remove the rows where `returned` is missing.
6. Remove the rows where `items` is below 1 or above 20. Keep the rows where it is missing.
7. Fill the missing values of `previous_orders` and of `size_chart_views` with the median of that column.
8. Merge with the lookup file on `city`, keeping every row of your table.
9. Add a column `flag` that is 1 where `returned` is `Yes` and 0 otherwise.

Check values: your program should print these

Rows in the raw file	836
Rows after step 1 (exact copies dropped)	820
Rows after step 5 (missing target removed)	808
Rows after step 6 (impossible `items` removed)	800
Median used to fill `previous_orders`	3.00
Median used to fill `size_chart_views`	1.00
Rows and columns of the final table	800 and 13
Missing values left in the final table	0
Share of rows with `flag` equal to 1	0.307
Part D: train rows and test rows (seed 2012)	626 and 174
Part D: baseline accuracy (always the more common label)	0.684
Part D: least-squares accuracy	0.747

Part C: Questions for This Task

C1. For each `city`, give the number of orders and the share of `returned` equal to `Yes` (the mean of `flag`). Which `city` has the highest value?

C2. Make a pivot table of the share of `returned` equal to `Yes` (the mean of `flag`), with `city` down the side and `category` across the top. Which pair of `city` and `category` gives the highest value?

C3. Group the rows by the month of `ordered_on`. Which month has the highest value of the share of `returned` equal to `Yes` (the mean of `flag`), and which has the lowest?

C4. Compare the share of `returned` equal to `Yes` (the mean of `flag`) across the groups of `province` from the lookup file. Then give the correlation between `courier_days` and `flag`, using `np.corrcoef`.

C5. Among shoe orders with a discount of 40 percent or more, what share came back?

Parts D and E: Names for This Task

Features for Part D, in this order: `items`, `discount_pct`, `previous_orders`, `size_chart_views`, `order_value`, `courier_days`. The value to predict is `flag`. Use the seed 2012 for the split.

For Part E, show `returned` with a count plot coloured by `category`. Compare the groups of `city` with a bar plot of the mean of `flag`. For the outlier check (E7), use your strongest feature from step D3.

Next step. In the final project you will train classification models that predict whether `returned` is `Yes` for a new order. They must beat the accuracies above.

Task 13: Electricity Used by Steel Furnaces in Gujranwala

Group 13 Target for the final project: `energy_kwh` (regression)

Gujranwala Melt Works runs induction furnaces at six units around the city. Each row is one batch of scrap melted in 2025. The company wants to plan its electricity bill before a batch starts. Shift supervisors typed the unit names. The data are invented for practice.

Files in the folder `Project_Phase1_Data`:

- data: `task13_gujranwala_furnaces.csv`, 530 rows
- lookup: `task13_gujranwala_furnaces_lookup.csv`, 6 rows

Columns of the data file

Column	Meaning	Unit or values
`batch_id`	label of the batch	text, one per row
`melted_on`	date of the melt	date, written YYYY-MM-DD
`unit`	unit that melted the batch	Kamoke, Eminabad, Sheikhupura Road, Grand Trunk Road, Wazirabad Road, Sialkot Bypass
`scrap_grade`	kind of scrap	Heavy, Light, Mixed
`scrap_tons`	weight of scrap in the batch	tons
`scrap_moisture_pct`	water in the scrap	percent
`ambient_temp_c`	air temperature outside	degrees C
`power_cuts`	power cuts during the batch	count
`scrap_price_per_ton`	price paid for the scrap	PKR, written as text such as Rs 45,000
`energy_kwh`	electricity used by the batch (target)	kWh

Columns of the lookup file

Column	Meaning	Unit or values
`unit`	unit that melted the batch	one row for each value
`furnace_capacity_tons`	largest batch the furnace can melt	tons
`power_source`	where the unit gets power	Generator, Grid, Grid and Generator

First lines of the data file

```
batch_id,melted_on,unit,scrap_grade,scrap_tons,scrap_moisture_pct,ambient_temp_c,power_cuts
    ,scrap_price_per_ton,energy_kwh
GS0001,2025-01-01,  SHEIKHUPURA ROAD ,Heavy,2.0,3.0,18,0,"Rs 133,000",1359
GS0002,2025-01-01,Sialkot Bypass,Mixed,7.26,3.4,19,5,"Rs 152,000",5802
GS0003,2025-01-04,Grand Trunk Road,Mixed,6.12,4.2,9,,"Rs 147,000",3737
```

Part B: Cleaning Rules for This Task

Apply these steps in this order. The check values below assume this order.

1. Drop rows that are exact copies of an earlier row.
2. Clean `unit`: remove spaces at both ends, then write each word with a capital first letter.
3. Turn `scrap_price_per_ton` from text such as `Rs 45,000` into numbers.
4. Turn `melted_on` into dates.
5. Remove the rows where `energy_kwh` is missing.

6. Remove the rows where `scrap_tons` is below 1 or above 20. Keep the rows where it is missing.
7. Fill the missing values of `scrap_moisture_pct` and of `power_cuts` with the median of that column.
8. Merge with the lookup file on `unit`, keeping every row of your table.

Check values: your program should print these

Rows in the raw file	530
Rows after step 1 (exact copies dropped)	520
Rows after step 5 (missing target removed)	512
Rows after step 6 (impossible `scrap_tons` removed)	507
Median used to fill `scrap_moisture_pct`	3.00
Median used to fill `power_cuts`	1.00
Rows and columns of the final table	507 and 12
Missing values left in the final table	0
Mean of `energy_kwh`	4,640.95
Part D: train rows and test rows (seed 2013)	405 and 102
Part D: baseline RMSE (predict the training mean)	1,325.74
Part D: least-squares RMSE	218.74

Part C: Questions for This Task

C1. For each `unit`, give the number of batches and the mean `energy_kwh`. Which `unit` has the highest value?

C2. Make a pivot table of the mean `energy_kwh`, with `unit` down the side and `scrap_grade` across the top. Which pair of `unit` and `scrap_grade` gives the highest value?

C3. Group the rows by the month of `melted_on`. Which month has the highest value of the mean `energy_kwh`, and which has the lowest?

C4. Compare the mean `energy_kwh` across the groups of `power_source` from the lookup file. Then give the correlation between `furnace_capacity_tons` and `energy_kwh`, using `np.corrcoef`.

C5. What is the mean energy per ton of scrap (energy_kwh divided by scrap_tons) for each scrap grade?

Parts D and E: Names for This Task

Features for Part D, in this order: `scrap_tons`, `scrap_moisture_pct`, `ambient_temp_c`, `power_cuts`, `scrap_price_per_ton`, `furnace_capacity_tons`. The value to predict is `energy_kwh`. Use the seed 2013 for the split.

For Part E, show `energy_kwh` with a histogram coloured by `scrap_grade`. Compare the groups of `unit` with a box plot. For the outlier check (E7), use `energy_kwh`.

Next step. In the final project you will train regression models that predict `energy_kwh` for a new batch. They must beat the RMSE values above.

Task 14: Cancelled Bookings at Northern Guesthouses

Group 14 Target for the final project: `cancelled` (classification)

Karakoram Stays takes bookings for guesthouses in seven hill towns. Each row is one booking for a stay in 2025. The company wants to know which bookings may be cancelled, so it can ask for a deposit. Guests type the town name themselves. The data are invented for practice.

Files in the folder `Project_Phase1_Data`:

- data: `task14_northern_guesthouses.csv`, 714 rows
- lookup: `task14_northern_guesthouses_lookup.csv`, 7 rows

Columns of the data file

Column	Meaning	Unit or values
`booking_id`	label of the booking	text, one per row
`stay_starts_on`	first night of the stay	date, written YYYY-MM-DD
`town`	town of the guesthouse	Karimabad, Naran, Kalam, Skardu, Khaplu, Chitral, Murree
`channel`	how the booking was made	Phone, Website, Walk In
`lead_days`	days between booking and arrival	days
`nights`	nights booked	count
`guests`	people in the booking	count
`previous_cancellations`	earlier bookings the guest cancelled	count
`booking_value`	total price of the stay	PKR, written as text such as Rs 45,000
`cancelled`	the booking was cancelled (target)	Yes or No

Columns of the lookup file

Column	Meaning	Unit or values
`town`	town of the guesthouse	one row for each value
`road_hours_from_islamabad`	hours by road from Islamabad	hours
`region`	province or region of the town	Gilgit Baltistan, Khyber Pakhtunkhwa, Punjab

First lines of the data file

```
booking_id,stay_starts_on,town,channel,lead_days,nights,guests,previous_cancellations,
    booking_value,cancelled
KS0001,2025-01-01,Naran,Website,19,3,1,0,"Rs 43,400",No
KS0002,2025-01-02,Murree,Phone,11,1,7,0,"Rs 40,900",No
KS0003,2025-01-02,Kalam,Website,9,5,8,0,"Rs 55,400",Yes
```

Part B: Cleaning Rules for This Task

Apply these steps in this order. The check values below assume this order.

1. Drop rows that are exact copies of an earlier row.
2. Clean `town`: remove spaces at both ends, then write each word with a capital first letter.
3. Turn `booking_value` from text such as `Rs 45,000` into numbers.
4. Turn `stay_starts_on` into dates.
5. Remove the rows where `cancelled` is missing.
6. Remove the rows where `nights` is below 1 or above 30. Keep the rows where it is missing.

7. Fill the missing values of `guests` and of `previous_cancellations` with the median of that column.
8. Merge with the lookup file on `town`, keeping every row of your table.
9. Add a column `flag` that is 1 where `cancelled` is `Yes` and 0 otherwise.

Check values: your program should print these

Rows in the raw file	714
Rows after step 1 (exact copies dropped)	700
Rows after step 5 (missing target removed)	690
Rows after step 6 (impossible `nights` removed)	683
Median used to fill `guests`	5.00
Median used to fill `previous_cancellations`	0.00
Rows and columns of the final table	683 and 13
Missing values left in the final table	0
Share of rows with `flag` equal to 1	0.325
Part D: train rows and test rows (seed 2014)	534 and 149
Part D: baseline accuracy (always the more common label)	0.664
Part D: least-squares accuracy	0.738

Part C: Questions for This Task

C1. For each `town`, give the number of bookings and the share of `cancelled` equal to `Yes` (the mean of `flag`). Which `town` has the highest value?

C2. Make a pivot table of the share of `cancelled` equal to `Yes` (the mean of `flag`), with `town` down the side and `channel` across the top. Which pair of `town` and `channel` gives the highest value?

C3. Group the rows by the month of `stay_starts_on`. Which month has the highest value of the share of `cancelled` equal to `Yes` (the mean of `flag`), and which has the lowest?

C4. Compare the share of `cancelled` equal to `Yes` (the mean of `flag`) across the groups of `region` from the lookup file. Then give the correlation between `road_hours_from_islamabad` and `flag`, using `np.corrcoef`.

C5. Among website bookings made 60 or more days ahead, what share were cancelled?

Parts D and E: Names for This Task

Features for Part D, in this order: `lead_days`, `nights`, `guests`, `previous_cancellations`, `booking_value`, `road_hours_from_islamabad`. The value to predict is `flag`. Use the seed 2014 for the split.

For Part E, show `cancelled` with a count plot coloured by `channel`. Compare the groups of `town` with a bar plot of the mean of `flag`. For the outlier check (E7), use your strongest feature from step D3.

Next step. In the final project you will train classification models that predict whether `cancelled` is `Yes` for a new booking. They must beat the accuracies above.

Task 15: Daily Output of Solar Systems in Bahawalpur

Group 15 Target for the final project: `daily_kwh` (regression)

Cholistan Solar Services installs and cleans rooftop and ground solar systems in and around Bahawalpur. Each row is one day of output from one system. The company wants to warn owners when a system makes less than it should. Technicians entered the site names on their phones. The data are invented for practice.

Files in the folder `Project_Phase1_Data`:

- data: `task15_bahawalpur_solar.csv`, 612 rows
- lookup: `task15_bahawalpur_solar_lookup.csv`, 7 rows

Columns of the data file

Column	Meaning	Unit or values
`reading_id`	label of the reading	text, one per row
`read_on`	date of the reading	date, written YYYY-MM-DD
`site`	area of the system	Model Town, Satellite Town, Baghdad Ul Jadeed, Yazman, Ahmadpur East, Hasilpur, Cantt
`mount`	where the panels are fixed	Roof, Ground
`system_kw`	size of the system	kW
`sunshine_hours`	hours of sunshine that day	hours
`days_since_cleaning`	days since the panels were washed	days
`temperature_c`	highest temperature that day	degrees C
`installation_cost`	price paid for the system	PKR, written as text such as Rs 45,000
`daily_kwh`	energy produced that day (target)	kWh

Columns of the lookup file

Column	Meaning	Unit or values
`site`	area of the system	one row for each value
`dust_storms_per_month`	usual dust storms in a month	count
`setting`	kind of area	City, Desert Edge, Town

First lines of the data file

```
reading_id,read_on,site,mount,system_kw,sunshine_hours,days_since_cleaning,temperature_c,
    installation_cost,daily_kwh
BS0001,2025-01-02,Yazman,Roof,3.0,9.5,5,38,"Rs 500,000",13.4
BS0002,2025-01-02,Yazman,Roof,10.0,5.6,45,39,"Rs 1,710,000",16.5
BS0003,2025-01-03,Baghdad Ul Jadeed,Ground,15.0,7.5,8,40,"Rs 2,690,000",
```

Part B: Cleaning Rules for This Task

Apply these steps in this order. The check values below assume this order.

1. Drop rows that are exact copies of an earlier row.
2. Clean `site`: remove spaces at both ends, then write each word with a capital first letter.
3. Turn `installation_cost` from text such as `Rs 45,000` into numbers.
4. Turn `read_on` into dates.
5. Remove the rows where `daily_kwh` is missing.

6. Remove the rows where `temperature_c` is below -5 or above 55. Keep the rows where it is missing.
7. Fill the missing values of `sunshine_hours` and of `days_since_cleaning` with the median of that column.
8. Merge with the lookup file on `site`, keeping every row of your table.

Check values: your program should print these

Rows in the raw file	612
Rows after step 1 (exact copies dropped)	600
Rows after step 5 (missing target removed)	591
Rows after step 6 (impossible `temperature_c` removed)	585
Median used to fill `sunshine_hours`	8.50
Median used to fill `days_since_cleaning`	22.50
Rows and columns of the final table	585 and 12
Missing values left in the final table	0
Mean of `daily_kwh`	29.85
Part D: train rows and test rows (seed 2015)	462 and 123
Part D: baseline RMSE (predict the training mean)	11.50
Part D: least-squares RMSE	3.11

Part C: Questions for This Task

C1. For each `site`, give the number of readings and the mean `daily_kwh`. Which `site` has the highest value?

C2. Make a pivot table of the mean `daily_kwh`, with `site` down the side and `mount` across the top. Which pair of `site` and `mount` gives the highest value?

C3. Group the rows by the month of `read_on`. Which month has the highest value of the mean `daily_kwh`, and which has the lowest?

C4. Compare the mean `daily_kwh` across the groups of `setting` from the lookup file. Then give the correlation between `dust_storms_per_month` and `daily_kwh`, using `np.corrcoef`.

C5. For 5 kW systems, how much more energy per day do panels washed within the last 7 days produce than panels not washed for 30 days or more?

Parts D and E: Names for This Task

Features for Part D, in this order: `system_kw`, `sunshine_hours`, `days_since_cleaning`, `temperature_c`, `installation_cost`, `dust_storms_per_month`. The value to predict is `daily_kwh`. Use the seed 2015 for the split.

For Part E, show `daily_kwh` with a histogram coloured by `mount`. Compare the groups of `site` with a box plot. For the outlier check (E7), use `daily_kwh`.

Next step. In the final project you will train regression models that predict `daily_kwh` for a new reading. They must beat the RMSE values above.

Task 16: Traffic at Motorway Toll Plazas

Group 16 Target for the final project: `vehicles_per_hour` (regression)

A toll operator on the Lahore to Islamabad motorway counts vehicles at eight plazas. Each row is one hour at one plaza in 2025. The operator wants to open the right number of lanes before the traffic arrives. Plaza names were typed by the staff on duty. The data are invented for practice.

Files in the folder `Project_Phase1_Data`:

- data: `task16_motorway_traffic.csv`, 775 rows
- lookup: `task16_motorway_traffic_lookup.csv`, 8 rows

Columns of the data file

Column	Meaning	Unit or values
`record_id`	label of the hourly count	text, one per row
`counted_on`	date of the count	date, written YYYY-MM-DD
`plaza`	toll plaza	Sheikhupura, Khanqah Dogran, Pindi Bhattian, Kot Momin, Bhera, Lilla, Kallar Kahar, Chakri
`day_type`	kind of day	Weekday, Weekend, Holiday
`hour`	hour of the day (0 to 23)	hour
`temperature_c`	air temperature	degrees C
`visibility_m`	how far drivers can see	metres
`heavy_vehicle_pct`	share of trucks and buses	percent
`average_toll`	average toll paid per vehicle	PKR, written as text such as Rs 45,000
`vehicles_per_hour`	vehicles that passed in the hour (target)	vehicles

Columns of the lookup file

Column	Meaning	Unit or values
`plaza`	toll plaza	one row for each value
`lanes`	toll lanes at the plaza	count
`district`	district of the plaza	Chakwal, Hafizabad, Jhelum, Rawalpindi, Sargodha, Sheikhupura

First lines of the data file

```
record_id,counted_on,plaza,day_type,hour,temperature_c,visibility_m,heavy_vehicle_pct,
    average_toll,vehicles_per_hour
MW0001,2025-01-01,Bhera,Weekend,21,24,8020,15.7,Rs 240,1217
MW0002,2025-01-02,Khanqah Dogran,Weekday,0,27,570,16.2,Rs 420,401
MW0003,2025-01-02,Pindi Bhattian,Weekend,21,22,5030,27.9,Rs 400,1561
```

Part B: Cleaning Rules for This Task

Apply these steps in this order. The check values below assume this order.

1. Drop rows that are exact copies of an earlier row.
2. Clean `plaza`: remove spaces at both ends, then write each word with a capital first letter.
3. Turn `average_toll` from text such as `Rs 45,000` into numbers.
4. Turn `counted_on` into dates.

5. Remove the rows where `vehicles_per_hour` is missing.
6. Remove the rows where `hour` is below 0 or above 23. Keep the rows where it is missing.
7. Fill the missing values of `visibility_m` and of `heavy_vehicle_pct` with the median of that column.
8. Merge with the lookup file on `plaza`, keeping every row of your table.

Check values: your program should print these

Rows in the raw file	775
Rows after step 1 (exact copies dropped)	760
Rows after step 5 (missing target removed)	749
Rows after step 6 (impossible `hour` removed)	741
Median used to fill `visibility_m`	3,100.00
Median used to fill `heavy_vehicle_pct`	21.90
Rows and columns of the final table	741 and 12
Missing values left in the final table	0
Mean of `vehicles_per_hour`	1,081.64
Part D: train rows and test rows (seed 2016)	594 and 147
Part D: baseline RMSE (predict the training mean)	388.22
Part D: least-squares RMSE	245.42

Part C: Questions for This Task

C1. For each `plaza`, give the number of hourly counts and the mean `vehicles_per_hour`. Which `plaza` has the highest value?

C2. Make a pivot table of the mean `vehicles_per_hour`, with `plaza` down the side and `day_type` across the top. Which pair of `plaza` and `day_type` gives the highest value?

C3. Group the rows by the month of `counted_on`. Which month has the highest value of the mean `vehicles_per_hour`, and which has the lowest?

C4. Compare the mean `vehicles_per_hour` across the groups of `district` from the lookup file. Then give the correlation between `lanes` and `vehicles_per_hour`, using `np.corrcoef`.

C5. On weekdays, which hour of the day has the highest mean traffic, and what is that mean?

Parts D and E: Names for This Task

Features for Part D, in this order: `hour`, `temperature_c`, `visibility_m`, `heavy_vehicle_pct`, `average_toll`, `lanes`. The value to predict is `vehicles_per_hour`. Use the seed 2016 for the split.

For Part E, show `vehicles_per_hour` with a histogram coloured by `day_type`. Compare the groups of `plaza` with a box plot. For the outlier check (E7), use `vehicles_per_hour`.

Next step. In the final project you will train regression models that predict `vehicles_per_hour` for a new hourly count. They must beat the RMSE values above.

Task 17: Milk Yield of Buffaloes in Pakpattan

Group 17 Target for the final project: `milk_litres` (regression)

Pakpattan Dairy Cooperative records the milk of buffaloes on member farms in six villages. Each row is one buffalo on one recording day in 2025. The cooperative wants to tell members how much milk to expect from an animal. Village names were written by the milk collectors. The data are invented for practice.

Files in the folder `Project_Phase1_Data`:

- data: `task17_pakpattan_milk.csv`, 632 rows
- lookup: `task17_pakpattan_milk_lookup.csv`, 6 rows

Columns of the data file

Column	Meaning	Unit or values
`record_id`	label of the record	text, one per row
`recorded_on`	date of the milk record	date, written YYYY-MM-DD
`village`	village of the farm	Arifwala, Noorpur, Qaboola, Kalyana, Chak Bedi, Malka Hans
`breed`	breed of the buffalo	Nili Ravi, Kundi, Crossbred
`age_years`	age of the buffalo	years
`feed_kg`	feed given per day	kg
`lactation_month`	months since the calf was born	months
`temperature_c`	highest temperature that day	degrees C
`feed_cost_per_day`	cost of the day's feed	PKR, written as text such as Rs 45,000
`milk_litres`	milk given that day (target)	litres

Columns of the lookup file

Column	Meaning	Unit or values
`village`	village of the farm	one row for each value
`vet_visits_per_month`	visits by the vet each month	count
`water_source`	main water for the animals	Canal, Tube Well

First lines of the data file

```
record_id,recorded_on,village,breed,age_years,feed_kg,lactation_month,temperature_c,
    feed_cost_per_day,milk_litres
PB0001,2025-01-01,Chak Bedi,Nili Ravi,5.7,18.3,7,38,"Rs 1,220",10.0
PB0002,2025-01-02,Arifwala,Nili Ravi,6.1,10.0,2,38,Rs 590,10.8
PB0003,2025-01-02,Qaboola,Nili Ravi,4.8,15.3,7,20,"Rs 1,000",10.9
```

Part B: Cleaning Rules for This Task

Apply these steps in this order. The check values below assume this order.

1. Drop rows that are exact copies of an earlier row.
2. Clean `village`: remove spaces at both ends, then write each word with a capital first letter.
3. Turn `feed_cost_per_day` from text such as `Rs 45,000` into numbers.
4. Turn `recorded_on` into dates.
5. Remove the rows where `milk_litres` is missing.
6. Remove the rows where `age_years` is below 2 or above 25. Keep the rows where it is missing.

7. Fill the missing values of `feed_kg` and of `temperature_c` with the median of that column.
8. Merge with the lookup file on `village`, keeping every row of your table.

Check values: your program should print these

Rows in the raw file	632
Rows after step 1 (exact copies dropped)	620
Rows after step 5 (missing target removed)	611
Rows after step 6 (impossible `age_years` removed)	605
Median used to fill `feed_kg`	13.80
Median used to fill `temperature_c`	28.00
Rows and columns of the final table	605 and 12
Missing values left in the final table	0
Mean of `milk_litres`	8.39
Part D: train rows and test rows (seed 2017)	480 and 125
Part D: baseline RMSE (predict the training mean)	3.39
Part D: least-squares RMSE	2.38

Part C: Questions for This Task

C1. For each `village`, give the number of records and the mean `milk_litres`. Which `village` has the highest value?

C2. Make a pivot table of the mean `milk_litres`, with `village` down the side and `breed` across the top. Which pair of `village` and `breed` gives the highest value?

C3. Group the rows by the month of `recorded_on`. Which month has the highest value of the mean `milk_litres`, and which has the lowest?

C4. Compare the mean `milk_litres` across the groups of `water_source` from the lookup file. Then give the correlation between `vet_visits_per_month` and `milk_litres`, using `np.corrcoef`.

C5. What is the mean daily milk of Nili Ravi buffaloes in their first three months of lactation?

Parts D and E: Names for This Task

Features for Part D, in this order: `age_years`, `feed_kg`, `lactation_month`, `temperature_c`, `feed_cost_per_day`, `vet_visits_per_month`. The value to predict is `milk_litres`. Use the seed 2017 for the split.

For Part E, show `milk_litres` with a histogram coloured by `breed`. Compare the groups of `village` with a box plot. For the outlier check (E7), use `milk_litres`.

Next step. In the final project you will train regression models that predict `milk_litres` for a new record. They must beat the RMSE values above.

Task 18: Delayed Flights out of Karachi

Group 18 Target for the final project: `delayed` (classification)

The operations desk of a Karachi airport tracks domestic departures to eight cities. Each row is one flight in 2025. The desk wants to warn passengers early when a flight is likely to leave late. Destination names were typed by several staff members. The data are invented for practice.

Files in the folder `Project_Phase1_Data`:

- data: `task18_karachi_flight_delays.csv`, 775 rows
- lookup: `task18_karachi_flight_delays_lookup.csv`, 8 rows

Columns of the data file

Column	Meaning	Unit or values
`flight_id`	label of the flight	text, one per row
`flew_on`	date of the flight	date, written YYYY-MM-DD
`destination`	city the flight goes to	Lahore, Islamabad, Peshawar, Quetta, Multan, Sukkur, Gwadar, Faisalabad
`carrier_type`	kind of airline	Full Service, Low Cost
`departure_hour`	planned hour of departure	hour
`aircraft_age_years`	age of the aircraft	years
`passengers`	passengers booked	count
`crosswind_kmh`	wind across the runway	km per hour
`average_fare`	average ticket price	PKR, written as text such as Rs 45,000
`delayed`	the flight left more than 30 minutes late (target)	Yes or No

Columns of the lookup file

Column	Meaning	Unit or values
`destination`	city the flight goes to	one row for each value
`flight_minutes`	usual time in the air	minutes
`route_type`	kind of route	Regional, Trunk

First lines of the data file

```
flight_id,flew_on,destination,carrier_type,departure_hour,aircraft_age_years,passengers,
    crosswind_kmh,average_fare,delayed
KF0001,2025-01-01,Peshawar,Low Cost,11,19.1,170,6,"Rs 21,900",No
KF0002,2025-01-01,Lahore,Full Service,13,17.8,143,11,"Rs 21,600",Yes
KF0003,2025-01-03,Lahore,Full Service,18,11.5,138,17,"Rs 28,200",Yes
```

Part B: Cleaning Rules for This Task

Apply these steps in this order. The check values below assume this order.

1. Drop rows that are exact copies of an earlier row.
2. Clean `destination`: remove spaces at both ends, then write each word with a capital first letter.
3. Turn `average_fare` from text such as `Rs 45,000` into numbers.
4. Turn `flew_on` into dates.
5. Remove the rows where `delayed` is missing.

6. Remove the rows where `passengers` is below 0 or above 200. Keep the rows where it is missing.
7. Fill the missing values of `aircraft_age_years` and of `crosswind_kmh` with the median of that column.
8. Merge with the lookup file on `destination`, keeping every row of your table.
9. Add a column `flag` that is 1 where `delayed` is `Yes` and 0 otherwise.

Check values: your program should print these

Rows in the raw file	775
Rows after step 1 (exact copies dropped)	760
Rows after step 5 (missing target removed)	749
Rows after step 6 (impossible `passengers` removed)	741
Median used to fill `aircraft_age_years`	11.30
Median used to fill `crosswind_kmh`	15.00
Rows and columns of the final table	741 and 13
Missing values left in the final table	0
Share of rows with `flag` equal to 1	0.402
Part D: train rows and test rows (seed 2018)	594 and 147
Part D: baseline accuracy (always the more common label)	0.592
Part D: least-squares accuracy	0.762

Part C: Questions for This Task

C1. For each `destination`, give the number of flights and the share of `delayed` equal to `Yes` (the mean of `flag`). Which `destination` has the highest value?

C2. Make a pivot table of the share of `delayed` equal to `Yes` (the mean of `flag`), with `destination` down the side and `carrier_type` across the top. Which pair of `destination` and `carrier_type` gives the highest value?

C3. Group the rows by the month of `flew_on`. Which month has the highest value of the share of `delayed` equal to `Yes` (the mean of `flag`), and which has the lowest?

C4. Compare the share of `delayed` equal to `Yes` (the mean of `flag`) across the groups of `route_type` from the lookup file. Then give the correlation between `flight_minutes` and `flag`, using `np.corrcoef`.

C5. Among evening departures (hour 17 or later) of low-cost airlines, what share left late?

Parts D and E: Names for This Task

Features for Part D, in this order: `departure_hour`, `aircraft_age_years`, `passengers`, `crosswind_kmh`, `average_fare`, `flight_minutes`. The value to predict is `flag`. Use the seed 2018 for the split.

For Part E, show `delayed` with a count plot coloured by `carrier_type`. Compare the groups of `destination` with a bar plot of the mean of `flag`. For the outlier check (E7), use your strongest feature from step D3.

Next step. In the final project you will train classification models that predict whether `delayed` is `Yes` for a new flight. They must beat the accuracies above.

Task 19: Resale Prices of Used Smartphones

Group 19 Target for the final project: `resale_price` (regression)

Second Chance Mobiles buys and resells used phones from stalls in six markets. Each row is one phone sold in 2025. The stalls want to quote a resale price the moment a phone is brought in. Stall keepers typed the market names. The data are invented for practice.

Files in the folder `Project_Phase1_Data`:

- data: `task19_used_smartphones.csv`, 714 rows
- lookup: `task19_used_smartphones_lookup.csv`, 6 rows

Columns of the data file

Column	Meaning	Unit or values
`phone_id`	label of the phone	text, one per row
`sold_on`	date of the sale	date, written YYYY-MM-DD
`market`	market of the stall	Hall Road, Hafeez Centre, Saddar, Chandni Chowk, Tariq Road, Gulberg
`has_box`	the phone comes with its box	Yes, No
`launch_price`	price of the phone when new	PKR (number)
`age_months`	months since first sale	months
`battery_health_pct`	battery capacity left	percent
`storage_gb`	storage	GB
`scratches`	visible scratches	count
`resale_price`	price the phone was sold for (target)	PKR, written as text such as Rs 45,000

Columns of the lookup file

Column	Meaning	Unit or values
`market`	market of the stall	one row for each value
`competing_shops`	phone shops in the market	count
`city`	city of the market	Karachi, Lahore, Rawalpindi

First lines of the data file

```
phone_id,sold_on,market,has_box,launch_price,age_months,battery_health_pct,storage_gb,
    scratches,resale_price
SC0001,2025-01-03,Gulberg,No,232000,6,80,128,4,"Rs 132,000"
SC0002,2025-01-03,Chandni Chowk,Yes,93000,12,82,256,2,
SC0003,2025-01-04,Chandni Chowk,No,73000,42,81,512,2,"Rs 28,500"
```

Part B: Cleaning Rules for This Task

Apply these steps in this order. The check values below assume this order.

1. Drop rows that are exact copies of an earlier row.
2. Clean `market`: remove spaces at both ends, then write each word with a capital first letter.
3. Turn `resale_price` from text such as `Rs 45,000` into numbers.
4. Turn `sold_on` into dates.
5. Remove the rows where `resale_price` is missing.
6. Remove the rows where `storage_gb` is below 16 or above 1,024. Keep the rows where it is missing.

7. Fill the missing values of `battery_health_pct` and of `scratches` with the median of that column.
8. Merge with the lookup file on `market`, keeping every row of your table.

Check values: your program should print these

Rows in the raw file	714
Rows after step 1 (exact copies dropped)	700
Rows after step 5 (missing target removed)	690
Rows after step 6 (impossible `storage_gb` removed)	683
Median used to fill `battery_health_pct`	86.00
Median used to fill `scratches`	1.00
Rows and columns of the final table	683 and 12
Missing values left in the final table	0
Mean of `resale_price`	53,728.55
Part D: train rows and test rows (seed 2019)	531 and 152
Part D: baseline RMSE (predict the training mean)	34,254.18
Part D: least-squares RMSE	5,175.86

Part C: Questions for This Task

C1. For each `market`, give the number of phones and the mean `resale_price`. Which `market` has the highest value?

C2. Make a pivot table of the mean `resale_price`, with `market` down the side and `has_box` across the top. Which pair of `market` and `has_box` gives the highest value?

C3. Group the rows by the month of `sold_on`. Which month has the highest value of the mean `resale_price`, and which has the lowest?

C4. Compare the mean `resale_price` across the groups of `city` from the lookup file. Then give the correlation between `competing_shops` and `resale_price`, using `np.corrcoef`.

C5. For phones older than 24 months, how much higher is the mean resale price when the box is included?

Parts D and E: Names for This Task

Features for Part D, in this order: `launch_price`, `age_months`, `battery_health_pct`, `storage_gb`, `scratches`, `competing_shops`. The value to predict is `resale_price`. Use the seed 2019 for the split.

For Part E, show `resale_price` with a histogram coloured by `has_box`. Compare the groups of `market` with a box plot. For the outlier check (E7), use `resale_price`.

Next step. In the final project you will train regression models that predict `resale_price` for a new phone. They must beat the RMSE values above.

Task 20: Staff Leaving a Lahore Software House

Group 20 Target for the final project: `left_company` (classification)

Ravi Code Labs is a software company in Lahore with six teams. Each row is one employee on the payroll at the start of 2025. Human resources wants to know who may resign, so managers can talk to them early. Team names were copied from several old spreadsheets. The data are invented for practice.

Files in the folder `Project_Phase1_Data`:

- data: `task20_software_house_attrition.csv`, 663 rows
- lookup: `task20_software_house_attrition_lookup.csv`, 6 rows

Columns of the data file

Column	Meaning	Unit or values
`employee_id`	label of the employee	text, one per row
`joined_on`	date the employee joined	date, written YYYY-MM-DD
`team`	team of the employee	Backend, Mobile, Quality Assurance, Design, Support, Data
`work_mode`	where the employee works	Office, Hybrid, Remote
`years_at_company`	years with the company	years
`overtime_hours_month`	overtime worked per month	hours
`commute_km`	distance from home to office	km
`training_days_year`	training days given last year	days
`monthly_salary`	salary per month	PKR, written as text such as Rs 45,000
`left_company`	the employee resigned during 2025 (target)	Yes or No

Columns of the lookup file

Column	Meaning	Unit or values
`team`	team of the employee	one row for each value
`team_size`	people in the team	count
`office`	office of the team	Gulberg Office, Johar Town Office, Model Town Office

First lines of the data file

```
employee_id,joined_on,team,work_mode,years_at_company,overtime_hours_month,commute_km,
    training_days_year,monthly_salary,left_company
RC0001,2018-01-03,Quality Assurance,Office,7.0,9,26.9,9,"Rs 105,000",No
RC0002,2018-01-09,Support,Office,7.0,12,20.5,12,"Rs 234,000",Yes
RC0003,2018-01-13,  BACKEND ,Remote,7.0,13,10.3,5,"Rs 183,000",No
```

Part B: Cleaning Rules for This Task

Apply these steps in this order. The check values below assume this order.

1. Drop rows that are exact copies of an earlier row.
2. Clean `team`: remove spaces at both ends, then write each word with a capital first letter.
3. Turn `monthly_salary` from text such as `Rs 45,000` into numbers.
4. Turn `joined_on` into dates.
5. Remove the rows where `left_company` is missing.

6. Remove the rows where `overtime_hours_month` is below 0 or above 200. Keep the rows where it is missing.
7. Fill the missing values of `commute_km` and of `training_days_year` with the median of that column.
8. Merge with the lookup file on `team`, keeping every row of your table.
9. Add a column `flag` that is 1 where `left_company` is `Yes` and 0 otherwise.

Check values: your program should print these

Rows in the raw file	663
Rows after step 1 (exact copies dropped)	650
Rows after step 5 (missing target removed)	640
Rows after step 6 (impossible `overtime_hours_month` removed)	634
Median used to fill `commute_km`	9.10
Median used to fill `training_days_year`	7.00
Rows and columns of the final table	634 and 13
Missing values left in the final table	0
Share of rows with `flag` equal to 1	0.246
Part D: train rows and test rows (seed 2020)	497 and 137
Part D: baseline accuracy (always the more common label)	0.781
Part D: least-squares accuracy	0.803

Part C: Questions for This Task

C1. For each `team`, give the number of employees and the share of `left_company` equal to `Yes` (the mean of `flag`). Which `team` has the highest value?

C2. Make a pivot table of the share of `left_company` equal to `Yes` (the mean of `flag`), with `team` down the side and `work_mode` across the top. Which pair of `team` and `work_mode` gives the highest value?

C3. Group the rows by the month of `joined_on`. Which month has the highest value of the share of `left_company` equal to `Yes` (the mean of `flag`), and which has the lowest?

C4. Compare the share of `left_company` equal to `Yes` (the mean of `flag`) across the groups of `office` from the lookup file. Then give the correlation between `team_size` and `flag`, using `np.corrcoef`.

C5. Among office-based staff with 18 or more overtime hours a month, what share resigned?

Parts D and E: Names for This Task

Features for Part D, in this order: `years_at_company`, `overtime_hours_month`, `commute_km`, `training_days_year`, `monthly_salary`, `team_size`. The value to predict is `flag`. Use the seed 2020 for the split.

For Part E, show `left_company` with a count plot coloured by `work_mode`. Compare the groups of `team` with a bar plot of the mean of `flag`. For the outlier check (E7), use your strongest feature from step D3.

Next step. In the final project you will train classification models that predict whether `left_company` is `Yes` for a new employee. They must beat the accuracies above.

Task 21: Sugar Recovery at Mills in Sindh

Group 21 Target for the final project: `recovery_pct` (regression)

Mehran Sugar Mills buys sugarcane from growers in seven areas of lower Sindh. Each row is one truckload of cane delivered in the 2024 to 2025 season. The mills want to predict how much sugar a load will give before it is crushed. Weighbridge clerks typed the area names. The data are invented for practice.

Files in the folder `Project_Phase1_Data`:

- data: `task21_sindh_sugar_recovery.csv`, 653 rows
- lookup: `task21_sindh_sugar_recovery_lookup.csv`, 7 rows

Columns of the data file

Column	Meaning	Unit or values
`delivery_id`	label of the delivery	text, one per row
`delivered_on`	date the cane reached the mill	date, written YYYY-MM-DD
`mill_area`	area where the cane was grown	Badin, Thatta, Matiari, Tando Allahyar, Mirpur Khas, Sanghar, Nawabshah
`variety`	maturity group of the cane	Early, Mid Season, Late
`cane_age_months`	age of the cane at harvest	months
`harvest_to_crush_hours`	hours between cutting and crushing	hours
`cane_brix`	sugar reading of the juice	degrees Brix
`trash_pct`	leaves and soil in the load	percent
`cane_price_per_maund`	price paid for the cane	PKR, written as text such as Rs 45,000
`recovery_pct`	sugar made per 100 kg of cane (target)	percent

Columns of the lookup file

Column	Meaning	Unit or values
`mill_area`	area where the cane was grown	one row for each value
`crushing_capacity_tcd`	cane the nearest mill can crush per day	tons per day
`soil`	main soil of the area	Loam, Saline, Sandy Loam

First lines of the data file

```
delivery_id,delivered_on,mill_area,variety,cane_age_months,harvest_to_crush_hours,cane_brix
    ,trash_pct,cane_price_per_maund,recovery_pct
SS0001,2024-11-01,Nawabshah,Early,12.0,67,19.9,7.7,Rs 413,9.54
SS0002,2024-11-01,Badin,Late,12.8,19,17.7,,Rs 420,8.96
SS0003,2024-11-01,Thatta,Late,10.7,29,20.9,7.5,Rs 427,9.63
```

Part B: Cleaning Rules for This Task

Apply these steps in this order. The check values below assume this order.

1. Drop rows that are exact copies of an earlier row.
2. Clean `mill_area`: remove spaces at both ends, then write each word with a capital first letter.
3. Turn `cane_price_per_maund` from text such as `Rs 45,000` into numbers.
4. Turn `delivered_on` into dates.

5. Remove the rows where `recovery_pct` is missing.
6. Remove the rows where `cane_age_months` is below 6 or above 20. Keep the rows where it is missing.
7. Fill the missing values of `cane_brix` and of `trash_pct` with the median of that column.
8. Merge with the lookup file on `mill_area`, keeping every row of your table.

Check values: your program should print these

Rows in the raw file	653
Rows after step 1 (exact copies dropped)	640
Rows after step 5 (missing target removed)	630
Rows after step 6 (impossible `cane_age_months` removed)	624
Median used to fill `cane_brix`	18.80
Median used to fill `trash_pct`	6.10
Rows and columns of the final table	624 and 12
Missing values left in the final table	0
Mean of `recovery_pct`	10.26
Part D: train rows and test rows (seed 2021)	494 and 130
Part D: baseline RMSE (predict the training mean)	0.94
Part D: least-squares RMSE	0.50

Part C: Questions for This Task

C1. For each `mill_area`, give the number of deliveries and the mean `recovery_pct`. Which `mill_area` has the highest value?

C2. Make a pivot table of the mean `recovery_pct`, with `mill_area` down the side and `variety` across the top. Which pair of `mill_area` and `variety` gives the highest value?

C3. Group the rows by the month of `delivered_on`. Which month has the highest value of the mean `recovery_pct`, and which has the lowest?

C4. Compare the mean `recovery_pct` across the groups of `soil` from the lookup file. Then give the correlation between `crushing_capacity_tcd` and `recovery_pct`, using `np.corrcoef`.

C5. What is the mean recovery of loads crushed within 24 hours of cutting, and of loads crushed after 48 hours or more?

Parts D and E: Names for This Task

Features for Part D, in this order: `cane_age_months`, `harvest_to_crush_hours`, `cane_brix`, `trash_pct`, `cane_price_per_maund`, `crushing_capacity_tcd`. The value to predict is `recovery_pct`. Use the seed 2021 for the split.

For Part E, show `recovery_pct` with a histogram coloured by `variety`. Compare the groups of `mill_area` with a box plot. For the outlier check (E7), use `recovery_pct`.

Next step. In the final project you will train regression models that predict `recovery_pct` for a new delivery. They must beat the RMSE values above.

Task 22: Children Leaving School in Tharparkar

Group 22 Target for the final project: `dropped_out` (classification)

Thar Learning Trust supports primary schools in seven talukas of Tharparkar. Each row is one child enrolled at the start of the 2025 school year. The trust wants to visit the families of children who may leave school, before they leave. Teachers wrote the taluka names on survey sheets. The data are invented for practice.

Files in the folder `Project_Phase1_Data`:

- data: `task22_thar_school_dropout.csv`, 612 rows
- lookup: `task22_thar_school_dropout_lookup.csv`, 7 rows

Columns of the data file

Column	Meaning	Unit or values
`enrolment_id`	label of the child	text, one per row
`surveyed_on`	date of the home survey	date, written YYYY-MM-DD
`taluka`	taluka of the school	Mithi, Islamkot, Chachro, Diplo, Nagarparkar, Dahli, Kaloi
`water_source`	where the family gets water	Piped, Well, Tanker
`distance_to_school_km`	distance from home to school	km
`household_size`	people in the household	count
`siblings_in_school`	brothers and sisters also in school	count
`days_absent_first_term`	days absent in the first term	days
`family_monthly_income`	income of the family per month	PKR, written as text such as Rs 45,000
`dropped_out`	the child left school before the year ended (target)	Yes or No

Columns of the lookup file

Column	Meaning	Unit or values
`taluka`	taluka of the school	one row for each value
`paved_road_km`	paved roads in the taluka	km
`terrain`	land of the taluka	Desert, Hills, Town

First lines of the data file

```
enrolment_id,surveyed_on,taluka,water_source,distance_to_school_km,household_size,
    siblings_in_school,days_absent_first_term,family_monthly_income,dropped_out
TL0001,2025-02-01,Nagarparkar,Well,3.0,4,3,13,"Rs 23,800",Yes
TL0002,2025-02-01,Nagarparkar,Piped,0.7,12,5,11,"Rs 40,100",Yes
TL0003,2025-02-01,Nagarparkar,Tanker,2.3,13,5,9,"Rs 26,200",Yes
```

Part B: Cleaning Rules for This Task

Apply these steps in this order. The check values below assume this order.

1. Drop rows that are exact copies of an earlier row.
2. Clean `taluka`: remove spaces at both ends, then write each word with a capital first letter.
3. Turn `family_monthly_income` from text such as `Rs 45,000` into numbers.
4. Turn `surveyed_on` into dates.
5. Remove the rows where `dropped_out` is missing.

6. Remove the rows where `household_size` is below 1 or above 25. Keep the rows where it is missing.
7. Fill the missing values of `siblings_in_school` and of `days_absent_first_term` with the median of that column.
8. Merge with the lookup file on `taluka`, keeping every row of your table.
9. Add a column `flag` that is 1 where `dropped_out` is `Yes` and 0 otherwise.

Check values: your program should print these

Rows in the raw file	612
Rows after step 1 (exact copies dropped)	600
Rows after step 5 (missing target removed)	591
Rows after step 6 (impossible `household_size` removed)	585
Median used to fill `siblings_in_school`	3.00
Median used to fill `days_absent_first_term`	8.00
Rows and columns of the final table	585 and 13
Missing values left in the final table	0
Share of rows with `flag` equal to 1	0.356
Part D: train rows and test rows (seed 2022)	475 and 110
Part D: baseline accuracy (always the more common label)	0.645
Part D: least-squares accuracy	0.755

Part C: Questions for This Task

C1. For each `taluka`, give the number of children and the share of `dropped_out` equal to `Yes` (the mean of `flag`). Which `taluka` has the highest value?

C2. Make a pivot table of the share of `dropped_out` equal to `Yes` (the mean of `flag`), with `taluka` down the side and `water_source` across the top. Which pair of `taluka` and `water_source` gives the highest value?

C3. Group the rows by the month of `surveyed_on`. Which month has the highest value of the share of `dropped_out` equal to `Yes` (the mean of `flag`), and which has the lowest?

C4. Compare the share of `dropped_out` equal to `Yes` (the mean of `flag`) across the groups of `terrain` from the lookup file. Then give the correlation between `paved_road_km` and `flag`, using `np.corrcoef`.

C5. Among children who live more than 5 km from school, what share left school?

Parts D and E: Names for This Task

Features for Part D, in this order: `distance_to_school_km`, `household_size`, `siblings_in_school`, `days_absent_first_term`, `family_monthly_income`, `paved_road_km`. The value to predict is `flag`. Use the seed 2022 for the split.

For Part E, show `dropped_out` with a count plot coloured by `water_source`. Compare the groups of `taluka` with a bar plot of the mean of `flag`. For the outlier check (E7), use your strongest feature from step D3.

Next step. In the final project you will train classification models that predict whether `dropped_out` is `Yes` for a new child. They must beat the accuracies above.

Task 23: Restaurant Ratings in Islamabad and Rawalpindi

Group 23 Target for the final project: `average_rating` (regression)

Capital Food Guide publishes reviews of restaurants in seven markets of Islamabad and Rawalpindi. Each row is one restaurant. The guide wants to estimate the rating a new restaurant is likely to earn. Reviewers typed the market names in the guide's app. The data are invented for practice.

Files in the folder `Project_Phase1_Data`:

- data: `task23_islamabad_restaurants.csv`, 571 rows
- lookup: `task23_islamabad_restaurants_lookup.csv`, 7 rows

Columns of the data file

Column	Meaning	Unit or values
`restaurant_id`	label of the restaurant	text, one per row
`last_reviewed_on`	date of the latest review	date, written YYYY-MM-DD
`market`	market of the restaurant	Jinnah Super, Kohsar Market, Melody Market, Super Market, Blue Area, Bahria Town, Saddar
`cuisine`	main kind of food	Desi, Chinese, Fast Food, Barbecue, Continental
`seats`	seats in the restaurant	count
`menu_items`	dishes on the menu	count
`avg_wait_min`	average wait for food	minutes
`years_open`	years since opening	years
`avg_bill_per_person`	average bill for one person	PKR, written as text such as Rs 45,000
`average_rating`	average review score (target)	1 to 5 stars

Columns of the lookup file

Column	Meaning	Unit or values
`market`	market of the restaurant	one row for each value
`parking_spaces`	parking spaces in the market	count
`city`	city of the market	Islamabad, Rawalpindi

First lines of the data file

```
restaurant_id,last_reviewed_on,market,cuisine,seats,menu_items,avg_wait_min,years_open,
    avg_bill_per_person,average_rating
CF0001,2025-01-01,Blue Area,Desi,24,44,32,3.3,"Rs 1,380",3.6
CF0002,2025-01-02,Bahria Town,Desi,92,28,32,2.3,"Rs 1,750",3.4
CF0003,2025-01-02,Saddar,Continental,76,46,8,3.1,"Rs 1,450",3.9
```

Part B: Cleaning Rules for This Task

Apply these steps in this order. The check values below assume this order.

1. Drop rows that are exact copies of an earlier row.
2. Clean `market`: remove spaces at both ends, then write each word with a capital first letter.
3. Turn `avg_bill_per_person` from text such as `Rs 45,000` into numbers.
4. Turn `last_reviewed_on` into dates.
5. Remove the rows where `average_rating` is missing.

6. Remove the rows where `avg_wait_min` is below 0 or above 180. Keep the rows where it is missing.
7. Fill the missing values of `menu_items` and of `years_open` with the median of that column.
8. Merge with the lookup file on `market`, keeping every row of your table.

Check values: your program should print these

Rows in the raw file	571
Rows after step 1 (exact copies dropped)	560
Rows after step 5 (missing target removed)	552
Rows after step 6 (impossible `avg_wait_min` removed)	546
Median used to fill `menu_items`	45.00
Median used to fill `years_open`	4.85
Rows and columns of the final table	546 and 12
Missing values left in the final table	0
Mean of `average_rating`	3.91
Part D: train rows and test rows (seed 2023)	438 and 108
Part D: baseline RMSE (predict the training mean)	0.44
Part D: least-squares RMSE	0.27

Part C: Questions for This Task

C1. For each `market`, give the number of restaurants and the mean `average_rating`. Which `market` has the highest value?

C2. Make a pivot table of the mean `average_rating`, with `market` down the side and `cuisine` across the top. Which pair of `market` and `cuisine` gives the highest value?

C3. Group the rows by the month of `last_reviewed_on`. Which month has the highest value of the mean `average_rating`, and which has the lowest?

C4. Compare the mean `average_rating` across the groups of `city` from the lookup file. Then give the correlation between `parking_spaces` and `average_rating`, using `np.corrcoef`.

C5. Among restaurants with an average wait under 15 minutes, which cuisine has the highest mean rating?

Parts D and E: Names for This Task

Features for Part D, in this order: `seats`, `menu_items`, `avg_wait_min`, `years_open`, `avg_bill_per_person`, `parking_spaces`. The value to predict is `average_rating`. Use the seed 2023 for the split.

For Part E, show `average_rating` with a histogram coloured by `cuisine`. Compare the groups of `market` with a box plot. For the outlier check (E7), use `average_rating`.

Next step. In the final project you will train regression models that predict `average_rating` for a new restaurant. They must beat the RMSE values above.

Task 24: Fraud on a Mobile Wallet in Sindh

Group 24 Target for the final project: `fraud` (classification)

Bandar Pay runs a mobile wallet used in seven cities of Sindh. Each row is one money transfer in 2025. The company wants to hold suspicious transfers for a check before the money leaves. City names come from the address customers typed when they signed up. The data are invented for practice.

Files in the folder `Project_Phase1_Data`:

- data: `task24_mobile_wallet_fraud.csv`, 918 rows
- lookup: `task24_mobile_wallet_fraud_lookup.csv`, 7 rows

Columns of the data file

Column	Meaning	Unit or values
`transaction_id`	label of the transfer	text, one per row
`sent_on`	date of the transfer	date, written YYYY-MM-DD
`city`	city of the sender	Karachi, Hyderabad, Sukkur, Larkana, Nawabshah, Mirpur Khas, Thatta
`channel`	how the transfer was made	App, USSD, Agent
`hour`	hour of the transfer (0 to 23)	hour
`account_age_days`	days since the account was opened	days
`transfers_last_24h`	transfers by the same account in the last day	count
`failed_pin_attempts`	wrong PIN entries before this transfer	count
`amount`	amount sent	PKR, written as text such as Rs 45,000
`fraud`	the transfer was confirmed as fraud (target)	Yes or No

Columns of the lookup file

Column	Meaning	Unit or values
`city`	city of the sender	one row for each value
`agents_registered`	wallet agents in the city	count
`region`	part of Sindh	Central Sindh, Coastal, Lower Sindh, Metro, Upper Sindh

First lines of the data file

```
transaction_id,sent_on,city,channel,hour,account_age_days,transfers_last_24h,
    failed_pin_attempts,amount,fraud
BP0001,2025-01-01,Karachi,App,12,584,4,0,"Rs 6,440",Yes
BP0002,2025-01-01,Karachi,App,1,226,3,0,"Rs 2,020",No
BP0003,2025-01-01,Karachi,Agent,22,474,4,0,"Rs 27,150",No
```

Part B: Cleaning Rules for This Task

Apply these steps in this order. The check values below assume this order.

1. Drop rows that are exact copies of an earlier row.
2. Clean `city`: remove spaces at both ends, then write each word with a capital first letter.
3. Turn `amount` from text such as `Rs 45,000` into numbers.
4. Turn `sent_on` into dates.

5. Remove the rows where `fraud` is missing.
6. Remove the rows where `hour` is below 0 or above 23. Keep the rows where it is missing.
7. Fill the missing values of `account_age_days` and of `failed_pin_attempts` with the median of that column.
8. Merge with the lookup file on `city`, keeping every row of your table.
9. Add a column `flag` that is 1 where `fraud` is `Yes` and 0 otherwise.

Check values: your program should print these

Rows in the raw file	918
Rows after step 1 (exact copies dropped)	900
Rows after step 5 (missing target removed)	886
Rows after step 6 (impossible `hour` removed)	877
Median used to fill `account_age_days`	399.00
Median used to fill `failed_pin_attempts`	0.00
Rows and columns of the final table	877 and 13
Missing values left in the final table	0
Share of rows with `flag` equal to 1	0.206
Part D: train rows and test rows (seed 2024)	688 and 189
Part D: baseline accuracy (always the more common label)	0.794
Part D: least-squares accuracy	0.810

Part C: Questions for This Task

C1. For each `city`, give the number of transfers and the share of `fraud` equal to `Yes` (the mean of `flag`). Which `city` has the highest value?

C2. Make a pivot table of the share of `fraud` equal to `Yes` (the mean of `flag`), with `city` down the side and `channel` across the top. Which pair of `city` and `channel` gives the highest value?

C3. Group the rows by the month of `sent_on`. Which month has the highest value of the share of `fraud` equal to `Yes` (the mean of `flag`), and which has the lowest?

C4. Compare the share of `fraud` equal to `Yes` (the mean of `flag`) across the groups of `region` from the lookup file. Then give the correlation between `agents_registered` and `flag`, using `np.corrcoef`.

C5. Among USSD transfers made between hour 0 and hour 4, what share were fraud?

Parts D and E: Names for This Task

Features for Part D, in this order: `hour`, `account_age_days`, `transfers_last_24h`, `failed_pin_attempts`, `amount`, `agents_registered`. The value to predict is `flag`. Use the seed 2024 for the split.

For Part E, show `fraud` with a count plot coloured by `channel`. Compare the groups of `city` with a bar plot of the mean of `flag`. For the outlier check (E7), use your strongest feature from step D3.

Next step. In the final project you will train classification models that predict whether `fraud` is `Yes` for a new transfer. They must beat the accuracies above.

Task 25: Groundwater Depth in Central Punjab

Group 25 Target for the final project: `water_depth_ft` (regression)

The Doab Groundwater Monitoring Project measures the water table at survey sites in eight tehsils. Each row is one measurement at one site in 2025. The project wants to estimate the depth at places it has not measured yet. Field teams wrote the tehsil names by hand. The data are invented for practice.

Files in the folder `Project_Phase1_Data`:

- data: `task25_punjab_groundwater.csv`, 551 rows
- lookup: `task25_punjab_groundwater_lookup.csv`, 8 rows

Columns of the data file

Column	Meaning	Unit or values
`site_id`	label of the measurement	text, one per row
`measured_on`	date of the measurement	date, written YYYY-MM-DD
`tehsil`	tehsil of the site	Kasur, Chunian, Pattoki, Depalpur, Renala Khurd, Okara, Sheikhupura, Ferozewala
`soil`	soil at the site	Sandy, Loam, Clay
`tubewells_per_sq_km`	tube wells per square km	count
`annual_rain_mm`	rain in the last year	mm
`distance_to_canal_km`	distance to the nearest canal	km
`rice_area_pct`	share of nearby land under rice	percent
`pumping_cost_per_hour`	cost of running a pump for an hour	PKR, written as text such as Rs 45,000
`water_depth_ft`	depth to the water table (target)	feet

Columns of the lookup file

Column	Meaning	Unit or values
`tehsil`	tehsil of the site	one row for each value
`canal_length_km`	canals in the tehsil	km
`doab`	land between two rivers	Bari, Rechna

First lines of the data file

```
site_id,measured_on,tehsil,soil,tubewells_per_sq_km,annual_rain_mm,distance_to_canal_km,
    rice_area_pct,pumping_cost_per_hour,water_depth_ft
DG0001,2025-01-02,Sheikhupura,Loam,5.0,355,19.6,24,Rs 950,77.3
DG0002,2025-01-03,Sheikhupura,Sandy,7.5,526,2.0,20,"Rs 1,330",50.1
DG0003,2025-01-03,Kasur,Clay,17.2,622,1.4,,Rs 790,48.6
```

Part B: Cleaning Rules for This Task

Apply these steps in this order. The check values below assume this order.

1. Drop rows that are exact copies of an earlier row.
2. Clean `tehsil`: remove spaces at both ends, then write each word with a capital first letter.
3. Turn `pumping_cost_per_hour` from text such as `Rs 45,000` into numbers.
4. Turn `measured_on` into dates.
5. Remove the rows where `water_depth_ft` is missing.

6. Remove the rows where `tubewells_per_sq_km` is below 0 or above 100. Keep the rows where it is missing.
7. Fill the missing values of `annual_rain_mm` and of `rice_area_pct` with the median of that column.
8. Merge with the lookup file on `tehsil`, keeping every row of your table.

Check values: your program should print these

Rows in the raw file	551
Rows after step 1 (exact copies dropped)	540
Rows after step 5 (missing target removed)	532
Rows after step 6 (impossible `tubewells_per_sq_km` removed)	527
Median used to fill `annual_rain_mm`	504.00
Median used to fill `rice_area_pct`	36.00
Rows and columns of the final table	527 and 12
Missing values left in the final table	0
Mean of `water_depth_ft`	61.02
Part D: train rows and test rows (seed 2025)	412 and 115
Part D: baseline RMSE (predict the training mean)	15.48
Part D: least-squares RMSE	6.69

Part C: Questions for This Task

C1. For each `tehsil`, give the number of measurements and the mean `water_depth_ft`. Which `tehsil` has the highest value?

C2. Make a pivot table of the mean `water_depth_ft`, with `tehsil` down the side and `soil` across the top. Which pair of `tehsil` and `soil` gives the highest value?

C3. Group the rows by the month of `measured_on`. Which month has the highest value of the mean `water_depth_ft`, and which has the lowest?

C4. Compare the mean `water_depth_ft` across the groups of `doab` from the lookup file. Then give the correlation between `canal_length_km` and `water_depth_ft`, using `np.corrcoef`.

C5. How much deeper, on average, is the water at sites more than 10 km from a canal than at sites within 2 km?

Parts D and E: Names for This Task

Features for Part D, in this order: `tubewells_per_sq_km`, `annual_rain_mm`, `distance_to_canal_km`, `rice_area_pct`, `pumping_cost_per_hour`, `canal_length_km`. The value to predict is `water_depth_ft`. Use the seed 2025 for the split.

For Part E, show `water_depth_ft` with a histogram coloured by `soil`. Compare the groups of `tehsil` with a box plot. For the outlier check (E7), use `water_depth_ft`.

Next step. In the final project you will train regression models that predict `water_depth_ft` for a new measurement. They must beat the RMSE values above.

Task 26: Gym Membership Renewals in Rawalpindi

Group 26 Target for the final project: `renewed` (classification)

Potohar Fitness Clubs runs six gyms in Rawalpindi. Each row is one membership that came up for renewal in 2025. The clubs want to know which members may not renew, so they can offer them a call or a free session. Front desk staff typed the branch names. The data are invented for practice.

Files in the folder `Project_Phase1_Data`:

- data: `task26_rawalpindi_gym_renewals.csv`, 653 rows
- lookup: `task26_rawalpindi_gym_renewals_lookup.csv`, 6 rows

Columns of the data file

Column	Meaning	Unit or values
`member_id`	label of the membership	text, one per row
`renewal_due_on`	date the renewal was due	date, written YYYY-MM-DD
`branch`	gym branch	Saddar, Satellite Town, Bahria Town, Gulraiz, Westridge, Chaklala
`plan`	length of the membership	Monthly, Quarterly, Yearly
`visits_per_month`	average visits per month	count
`age`	age of the member	years
`months_as_member`	months since first joining	months
`trainer_sessions`	sessions with a trainer last term	count
`monthly_fee`	fee per month	PKR, written as text such as Rs 45,000
`renewed`	the member renewed (target)	Yes or No

Columns of the lookup file

Column	Meaning	Unit or values
`branch`	gym branch	one row for each value
`machines`	exercise machines at the branch	count
`has_pool`	the branch has a pool	No, Yes

First lines of the data file

```
member_id,renewal_due_on,branch,plan,visits_per_month,age,months_as_member,trainer_sessions
    ,monthly_fee,renewed
PF0001,2025-01-01,Westridge,Quarterly,13,47,12,2,"Rs 4,200",Yes
PF0002,2025-01-02,Satellite Town,Monthly,-2,32,20,1,"Rs 7,500",Yes
PF0003,2025-01-02,Chaklala,Quarterly,11,25,21,4,"Rs 4,600",Yes
```

Part B: Cleaning Rules for This Task

Apply these steps in this order. The check values below assume this order.

1. Drop rows that are exact copies of an earlier row.
2. Clean `branch`: remove spaces at both ends, then write each word with a capital first letter.
3. Turn `monthly_fee` from text such as `Rs 45,000` into numbers.
4. Turn `renewal_due_on` into dates.
5. Remove the rows where `renewed` is missing.
6. Remove the rows where `visits_per_month` is below 0 or above 31. Keep the rows where it is missing.

7. Fill the missing values of `age` and of `trainer_sessions` with the median of that column.
8. Merge with the lookup file on `branch`, keeping every row of your table.
9. Add a column `flag` that is 1 where `renewed` is `Yes` and 0 otherwise.

Check values: your program should print these

Rows in the raw file	653
Rows after step 1 (exact copies dropped)	640
Rows after step 5 (missing target removed)	630
Rows after step 6 (impossible `visits_per_month` removed)	624
Median used to fill `age`	29.00
Median used to fill `trainer_sessions`	2.00
Rows and columns of the final table	624 and 13
Missing values left in the final table	0
Share of rows with `flag` equal to 1	0.660
Part D: train rows and test rows (seed 2026)	501 and 123
Part D: baseline accuracy (always the more common label)	0.707
Part D: least-squares accuracy	0.732

Part C: Questions for This Task

C1. For each `branch`, give the number of memberships and the share of `renewed` equal to `Yes` (the mean of `flag`). Which `branch` has the highest value?

C2. Make a pivot table of the share of `renewed` equal to `Yes` (the mean of `flag`), with `branch` down the side and `plan` across the top. Which pair of `branch` and `plan` gives the highest value?

C3. Group the rows by the month of `renewal_due_on`. Which month has the highest value of the share of `renewed` equal to `Yes` (the mean of `flag`), and which has the lowest?

C4. Compare the share of `renewed` equal to `Yes` (the mean of `flag`) across the groups of `has_pool` from the lookup file. Then give the correlation between `machines` and `flag`, using `np.corrcoef`.

C5. Among monthly-plan members who visit fewer than 6 times a month, what share renewed?

Parts D and E: Names for This Task

Features for Part D, in this order: `visits_per_month`, `age`, `months_as_member`, `trainer_sessions`, `monthly_fee`, `machines`. The value to predict is `flag`. Use the seed 2026 for the split.

For Part E, show `renewed` with a count plot coloured by `plan`. Compare the groups of `branch` with a bar plot of the mean of `flag`. For the outlier check (E7), use your strongest feature from step D3.

Next step. In the final project you will train classification models that predict whether `renewed` is `Yes` for a new membership. They must beat the accuracies above.

Task 27: Container Clearance Time at Karachi Port

Group 27 Target for the final project: `clearance_hours` (regression)

Keamari Clearing Agents clears import containers through five terminals at Karachi port. Each row is one container cleared in 2025. The agents want to tell importers how long their goods will take to leave the port. Terminal names were typed by different desk clerks. The data are invented for practice.

Files in the folder `Project_Phase1_Data`:

- data: `task27_karachi_port_clearance.csv`, 612 rows
- lookup: `task27_karachi_port_clearance_lookup.csv`, 5 rows

Columns of the data file

Column	Meaning	Unit or values
`container_id`	label of the container	text, one per row
`arrived_on`	date the container was unloaded	date, written YYYY-MM-DD
`terminal`	terminal that handled the container	East Wharf, West Wharf, South Terminal, Container Yard, Bulk Berth
`cargo_type`	kind of goods	Textile, Machinery, Food, Chemicals, Electronics
`weight_tons`	weight of the container	tons
`documents_missing`	documents missing at arrival	count
`inspection_queue`	containers waiting for inspection	count
`vessel_delay_hours`	hours the ship arrived late	hours
`declared_value`	value of the goods declared to customs	PKR, written as text such as Rs 45,000
`clearance_hours`	hours from unloading to leaving the port (target)	hours

Columns of the lookup file

Column	Meaning	Unit or values
`terminal`	terminal that handled the container	one row for each value
`cranes`	cranes at the terminal	count
`operator_type`	who runs the terminal	Private, Public

First lines of the data file

```
container_id,arrived_on,terminal,cargo_type,weight_tons,documents_missing,inspection_queue,
    vessel_delay_hours,declared_value,clearance_hours
KC0001,2025-01-01,East Wharf,Chemicals,25.3,0,23,6,"Rs 5,440,000",57
KC0002,2025-01-02,East Wharf,Electronics,19.4,3,30,6,"Rs 4,320,000",98
KC0003,2025-01-02,East Wharf,Electronics,16.3,0,0,2,"Rs 5,610,000",31
```

Part B: Cleaning Rules for This Task

Apply these steps in this order. The check values below assume this order.

1. Drop rows that are exact copies of an earlier row.
2. Clean `terminal`: remove spaces at both ends, then write each word with a capital first letter.
3. Turn `declared_value` from text such as `Rs 45,000` into numbers.
4. Turn `arrived_on` into dates.
5. Remove the rows where `clearance_hours` is missing.

6. Remove the rows where `weight_tons` is below 0.5 or above 35. Keep the rows where it is missing.
7. Fill the missing values of `inspection_queue` and of `vessel_delay_hours` with the median of that column.
8. Merge with the lookup file on `terminal`, keeping every row of your table.

Check values: your program should print these

Rows in the raw file	612
Rows after step 1 (exact copies dropped)	600
Rows after step 5 (missing target removed)	591
Rows after step 6 (impossible `weight_tons` removed)	585
Median used to fill `inspection_queue`	21.00
Median used to fill `vessel_delay_hours`	6.00
Rows and columns of the final table	585 and 12
Missing values left in the final table	0
Mean of `clearance_hours`	57.22
Part D: train rows and test rows (seed 2027)	467 and 118
Part D: baseline RMSE (predict the training mean)	24.86
Part D: least-squares RMSE	8.69

Part C: Questions for This Task

C1. For each `terminal`, give the number of containers and the mean `clearance_hours`. Which `terminal` has the highest value?

C2. Make a pivot table of the mean `clearance_hours`, with `terminal` down the side and `cargo_type` across the top. Which pair of `terminal` and `cargo_type` gives the highest value?

C3. Group the rows by the month of `arrived_on`. Which month has the highest value of the mean `clearance_hours`, and which has the lowest?

C4. Compare the mean `clearance_hours` across the groups of `operator_type` from the lookup file. Then give the correlation between `cranes` and `clearance_hours`, using `np.corrcoef`.

C5. For containers with no missing documents, which cargo type clears fastest on average?

Parts D and E: Names for This Task

Features for Part D, in this order: `weight_tons`, `documents_missing`, `inspection_queue`, `vessel_delay_hours`, `declared_value`, `cranes`. The value to predict is `clearance_hours`. Use the seed 2027 for the split.

For Part E, show `clearance_hours` with a histogram coloured by `cargo_type`. Compare the groups of `terminal` with a box plot. For the outlier check (E7), use `clearance_hours`.

Next step. In the final project you will train regression models that predict `clearance_hours` for a new container. They must beat the RMSE values above.